

Introduction to Business Processes

LEARNING OBJECTIVES

After completing this chapter you will be able to:

1. Define the functional organizational structure, and explain why this structure creates problems for modern businesses.
2. Describe key business processes in an organization.
3. Identify the main integration points between and among processes.
4. Understand the cross-functional nature of processes and their relationship to organizational areas.
5. Adopt and apply an integrated perspective to business processes.
6. Describe organizational structure of Global Bike Incorporated.
7. Explain how the SAP® ERP system promotes an integrated approach to business processes.

At this point in your university career, you have probably begun taking courses in accounting, operations, MIS, and other disciplines. These courses have introduced you to some basic business concepts and exposed you to different aspects of how a business operates. You have also begun to master several technology tools that will be very useful in your future career, such as office productivity tools to create spreadsheets, presentations, and documents. The course for which you are using this textbook deals with *integrated business processes* and the *enterprise systems* (ES) that support them. The concepts and skills you will gain from this course are different from what you have experienced previously, and you will need to approach this course with a different perspective.

■ THE FUNCTIONAL ORGANIZATIONAL STRUCTURE

To successfully master the concepts in this textbook, you must first begin to think holistically about the operations of a business. The most common organizational structure you are likely to encounter is the **functional structure**. Organizations that utilize a functional structure are divided into functions, or departments, each of which is responsible for a set of closely related activities. For example, the accounting department sends and receives payments, and the warehouse receives and ships materials. Typical functions or departments found in a modern organization include *purchasing, operations, warehouse, sales and marketing, research and development, finance and accounting, human resources, and information systems*. The vertical columns in Figure 1-1 identify the key functions in a typical company.

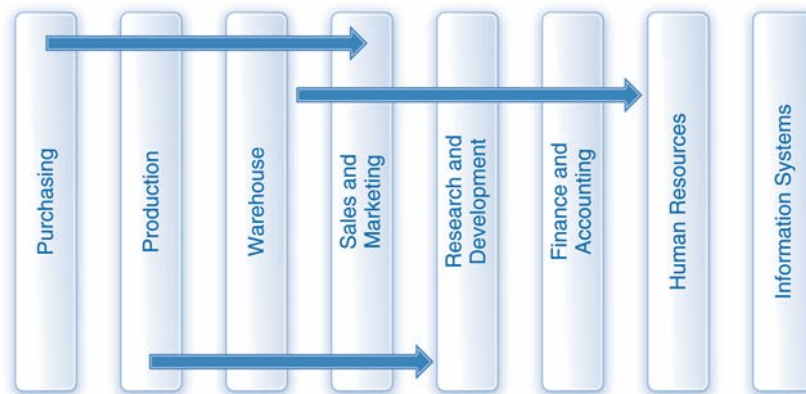


Figure 1-1: The functional structure

Although most companies maintain vertical (or functional) silos to compartmentalize their operational units, the integrated business processes that companies use to perform their work cut across these silos horizontally. Business processes, such as the procurement and fulfillment processes discussed later in the chapter, consist of activities that occur in different, seemingly unrelated functions or departments. In other words, these processes are *cross-functional*, meaning no single group or function is responsible for their execution. Rather, it is a shared responsibility among many functional areas. The cross-functional nature of business processes is also illustrated in Figure 1-1. For a process to be successfully completed, then, the company must rely on each functional group to execute its individual steps in the process in a coordinated way, which, as we shall see, may not be an easy thing to accomplish.

THE SILO EFFECT

The functional structure served organizations well for a number of years because it enabled them to cope with the challenges generated by their rapid growth. Over time, however, this system developed a serious drawback. Put

simply, people in the different functional areas came to perform their steps in the process in isolation, without fully understanding which steps happen before and which steps happen next. They essentially complete their part of the process, hand it off to the next person, and then proceed to the next task. By focusing so narrowly on their specific tasks, they lose sight of the “big picture” of the larger process, be it procurement, fulfillment, or any number of other common business processes. This tendency is commonly referred to as the **silo effect** because workers complete their tasks in their functional “silos” without regard to the consequences for the other components in the process.

A key point here is that the silo nature of the functional organizational structure and the cross-functional nature of processes are at odds with each other. That is, while workers focus on their specific function, each business process involves workers located in multiple functional areas. A major challenge facing organizations, then, is to coordinate activities among the different functional areas. Viewing a company from a process perspective requires employees to “think sideways”—in other words, to view the business across functional boundaries and focus on the end-to-end nature of the process and its intended outcomes. Learning to view a process from end to end is essential to understanding how enterprise systems help businesses manage their processes efficiently. Not surprisingly, then, this understanding has become a critical skill that companies have come to demand from their employees.

ENTERPRISE SYSTEMS

As you can see from the previous section, business processes span different parts of an organization. In fact, in today’s global economy, the various process steps are increasingly executed by people in multiple locations throughout the world. That is, a company will manufacture its products in different countries, acquire the materials to make these products from different locations, sell the products in many countries, and so on. For example, a bicycle manufacturer may purchase components from Italy, produce bicycles in Germany, and sell those bicycles in the United States. Because the steps in business processes are performed in locations that are geographically dispersed, it is impossible to manage such processes effectively without the use of modern information systems. Systems that support end-to-end processes are called **enterprise systems (ES)**, and they are essential to the efficient and effective execution and management of business process.

Given the significant impact that enterprise systems have on operational efficiency (and, ultimately, profitability), companies have invested enormous sums of capital and effort to plan, implement, and continuously improve enterprise systems over the past 40-plus years. A great deal of research has confirmed that investments in information technology (IT), particularly enterprise systems, have significantly increased the profitability, productivity, and competitiveness of corporations by removing the barriers to sharing information between functional areas and managing processes holistically.^{1,2} The key driver for this productivity and efficiency is the ability of modern enterprise systems

¹A. McAfee and E. Brynjolfsson, “Investing in the IT That Makes a Competitive Difference,” *Harvard Business Review*, 86, No. 7/8 (2008): 98–107.

²E. Brynjolfsson and L. Hitt, “Paradox Lost? Firm-level Evidence on the Returns to Information Systems Spending,” *Management Science*, 42, No. 4 (1996): 541–558.

to effectively manage a business process from beginning to end in an integrated, consistent, and highly efficient manner. Further, once a business process is managed by an integrated enterprise system, it can be monitored and improved very easily. As a result, we cannot discuss contemporary business processes without considering the role of enterprise systems. We will discuss enterprise systems in greater depth in Chapter 2.

In this chapter, we begin by identifying the key processes that typically exist in organizations. We then discuss enterprise systems and SAP, the company that produces the most popular enterprise systems globally. We also introduce Global Bike Incorporated, a company that we will use as a case study throughout this textbook to illustrate important concepts in a practical format. We conclude with the plan for the remainder of the book.

■ BUSINESS PROCESSES

Organizations exist either to serve some commercial purpose or to achieve some social objective. They differ depending on the purpose or goal they are trying to achieve, their ownership or management structure, and the regulatory environment in which they operate. Some organizations create and deliver products or services to customers to make a profit. For example, a bicycle manufacturer produces a variety of bicycles and accessories. It then sells these products to numerous retailers who, in turn, sell them to the final consumers. Other companies provide services, such as repairs to the bicycles. Yet others provide the manufacturer with the parts and materials needed to make the bicycles. Achieving the organization's objectives involves many different types of work. For example, the manufacturer must design the bicycles, identify what parts it will use to make them, determine where to obtain these parts, produce the bicycles, identify its customers, and market and sell the bicycles to them. In addition, it must determine how to manage its money, its various facilities such as factories and warehouses, and the many people that it must recruit, employ, train, and retain. This work is completed in numerous processes.

Although organizations exist for many different purposes, vary greatly in size and complexity, and operate in many different industries, they all exhibit similarities in the ways that they operate. Regardless of their type or size, successful organizations and industries use *processes* and *enterprise systems* to complete the work needed to achieve their goals. Processes may vary slightly depending on the unique characteristics of the industry or the structure of the organization, but the basic activities can be recognized by anyone who has developed a process view of business. Likewise, companies may employ different enterprise systems to manage their processes. However, you can apply the principles, concepts, and techniques explained in this textbook to most of the enterprise systems you are likely to work with.

A **business process**, illustrated in Figure 1-2, is a set of tasks or activities that produce desired outcomes. Every process is triggered by some event, such as receiving a customer order or recognizing the need to increase inventory. The columns in the figure represent different parts, or functional areas, within an organization, such as sales, warehouse, manufacturing, and accounting. Thus, the specific steps in the process are completed in different functional areas. For example, when a retailer (customer) places an order for bicycles, the manufacturer (seller) uses a specific process to ensure that the correct

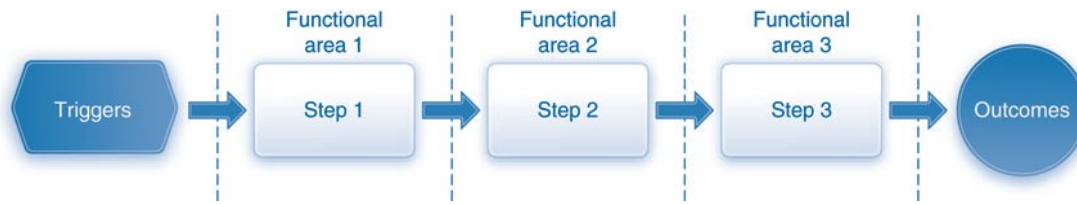


Figure 1-2: A generic business process

products are shipped to the customer in a timely manner and that payment for the order is received. These process steps can include validating the order, preparing the shipment, sending the shipment, issuing an invoice, and recording the receipt of payment. The sales department receives and validates the customer order and passes it on to the warehouse, which prepares and ships the order. The accounting department handles the invoice and payment steps. This is a very simplistic example. However, it highlights the fact that processes consist of interdependent steps that are completed in different parts of the organization.

Because the various process steps are carried out by different functional areas or departments, effective communication and collaboration among the departments is essential to the smooth execution of these processes. Without this interaction, the process cannot be completed efficiently and effectively. For instance, if the customer order is not properly communicated to the warehouse, then it cannot be shipped on time. Similarly, if the order and shipment information is not communicated to the accounting department, billing and payment will not be completed efficiently and accurately. Clearly, completing a process successfully requires more than just communicating information. Close coordination of work among the people involved is also essential. For example, when the salesperson accepts the order, he or she must collaborate with the warehouse to determine when the order can be shipped. Without this collaboration, the salesperson may make promises that the company cannot realistically meet. If this occurs, then, the products will not be available when promised. The salesperson must also collaborate with the accounting department to verify that the customer is credit-worthy. Accepting orders and shipping goods to customers who have not made payments for previous shipments can cause major financial problems for the organization.

An organization uses many processes to achieve its objectives, as illustrated in Figure 1-3. Three processes are directly related to creating and delivering products and services. They are *buy*, *make*, and *sell*. Organizations use specific terms to identify these processes.

- The **procurement process** (*buy*) refers to all of the activities involved in buying or acquiring the materials used by the organization, such as raw materials needed to make products.
- The **production process** (*make*) involves the actual creation of the products within the organization. Whereas the production process is concerned with acquiring needed materials *internally* (by making them), the procurement process is concerned with obtaining needed materials *externally* (by buying them). Each is appropriate for different types of materials, as we will discuss later in the book.

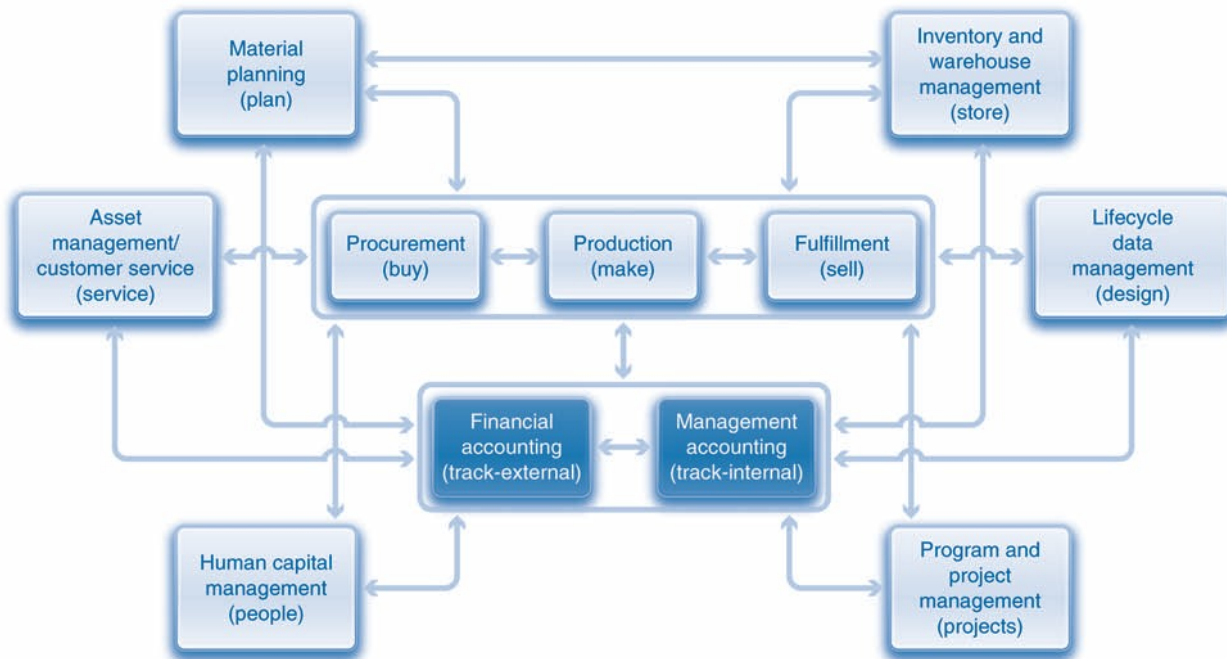


Figure 1-3: Key business processes

- Finally, the **fulfillment process** (*sell*) consists of all the steps involved in selling and delivering the products to the organization's customers.

Closely related to buying, making, and selling are four processes used to *design, plan, store, and service* products. Once again, organizations use specific terms for these processes.

- The **lifecycle data management process** (*design*) supports the design and development of products from the initial product idea stage through the discontinuation of the product.
- The **material planning process** (*plan*) uses historical data and sales forecasts to plan which materials will be procured and produced and in what quantities.
- The **inventory and warehouse management (IWM) process** (*store*) is used to store and track the materials.
- The **asset management and customer service processes** (*service*) are used to maintain internal assets such as machinery and to deliver after-sales customer service such as repairs.

Going further, two support processes are related to *people* and *projects*.

- **Human capital management (HCM) processes** (*people*) focus on the people within the organization and include functions such as recruiting, hiring, training, and benefits management.

- **Project management processes** (*projects*) are used to plan and execute large projects such as the construction of a new factory or the production of complex products such as airplanes.

All these processes have an impact on an organization's *finance*. This brings us to the last two processes, which *track* the financial impacts of processes.

- **Financial accounting (FI) processes** (*track-external*) track the financial impacts of process steps with the goal of meeting legal reporting requirements—for example, the Internal Revenue Service (IRS) or the Securities and Exchange Commission (SEC).
- **Management accounting or controlling (CO) processes** (*track-internal*) focus on internal reporting to manage costs and revenues.

Each of these processes can include numerous subprocesses. For example, each of the components of HCM, such as recruiting and benefits management, is itself a process. Similarly, IWM can include complex processes for receiving materials from a vendor and shipping products to a customer. In addition, each process can impact other processes, as illustrated by the arrows between the processes in Figure 1-3. These arrows represent process integration. For example, the procurement of raw materials has an impact on what can be produced and when. Similarly, the production process has an impact on what goods are available to sell and when. Going further, the arrows indicate that all processes have an impact on the organization's financials, a concept we explore throughout this book.

Clearly, then, in addition to understanding the details of how each process works, it is essential to understand the interrelationships among the processes. Significantly, to prevent Figure 1-3 from becoming cluttered with arrows, we did not include every possible integration point. Instead, we highlighted only the key points.

In the next section, we briefly describe the various business processes. We subsequently consider each process at length in separate chapters, where we also examine the linkages among the processes.

PROCUREMENT—BUY

The procurement process includes all of the tasks involved in *acquiring needed materials externally* from a vendor. A very simple example of a procurement process is diagrammed in Figure 1-4. As the figure illustrates, procurement is comprised of five steps that are completed in three different functional areas of the organization.

The process begins when the warehouse recognizes the need to procure materials, perhaps due to low levels of inventory. The warehouse then documents this need in the form of a purchase requisition, which it sends to the purchasing department. In turn, the purchasing department identifies a suitable vendor, creates a purchase order, and sends it to the vendor. The vendor ships the materials, which are received in the warehouse. The vendor then sends an invoice, which is received by the accounting department. Accounting then sends payment to the vendor, thereby completing the process.



Figure 1-4: A procurement process

In the preceding discussion a low inventory of materials was the trigger for the process. This discussion illustrates the link between procurement and the inventory and warehouse management process illustrated in Figure 1-3. Figure 1-3 shows, however, that procurement could be triggered by activity in other processes as well. The figure suggests at least three alternative scenarios.

1. The material planning process could indicate that the company needs to procure materials based on a forecasted demand for products.
2. The asset management or customer service process could trigger the procurement of a part needed to repair a machine or a product previously purchased by a customer.
3. A customer order (fulfillment process) could trigger the need to buy something, such as raw materials or component parts needed to manufacture the product.

PRODUCTION—MAKE

In the preceding discussion the company met the need that triggered the process via external procurement; that is, it purchased the needed materials from a vendor. Other times, however, a company uses the production process to *acquire needed materials internally*. As we explained in the previous paragraph, a customer order can trigger the production process. Alternatively, the material planning process can trigger in-house production. Figure 1-5 illustrates the case where the warehouse notices that its inventory of products is low. Subsequently, it will request production. In turn, the production department will approve the request. The approval authorizes the warehouse to release the materials needed to complete production. Once the production department has completed its task, the warehouse places the finished goods into storage. Note that this last step in the production process, which is concerned with the storage of the finished goods, could trigger IWM processes.

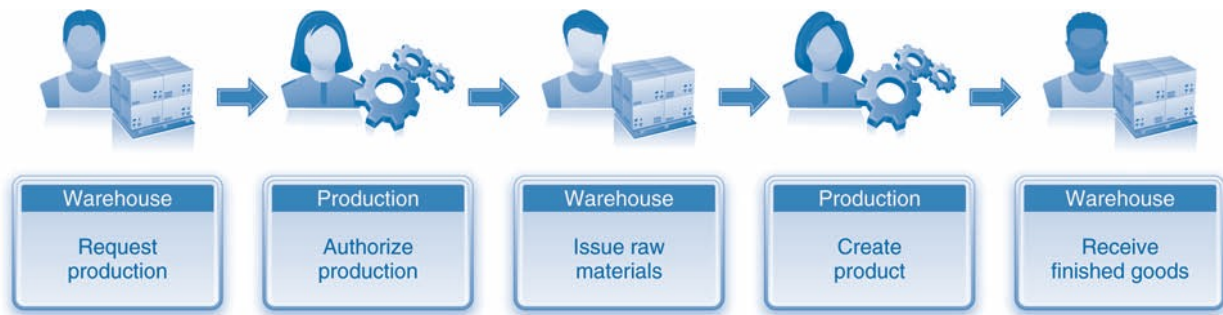


Figure 1-5: A production process

FULFILLMENT—SELL

Fulfillment (Figure 1-6) is concerned with *efficiently processing customer orders*. It is triggered by a customer purchase order that is received by the sales department. Sales then validates the order and creates a sales order. The sales order communicates data related to the order to other parts of the organization, and it tracks the progress of the order. The warehouse prepares and sends the shipment to the customer. Once accounting is notified of the shipment, it creates an invoice and sends it to the customer. The customer then makes a payment, which accounting records.

As this scenario illustrates, fulfillment triggers processes in IWM where the materials are stored. Of course, in many cases the ordered materials are not available in the warehouse. In such cases fulfillment will trigger external procurement and/or production.



Figure 1-6: A fulfillment process

MATERIAL PLANNING—PLAN

The term **material** encompasses all the products, components, parts, and so on that an organization uses. Businesses use and produce many types of materials. For example, material planning in a bicycle manufacturer would include: (a) *finished goods*, such as bicycles, that are sold to customers; (b) *semifinished goods*, such as wheel assemblies, that are used to make the finished goods; and

(c) *raw materials*, such as the tires, tubes, and wheels that are used to make the wheel assemblies. We examine the major *material types* in greater detail in Chapter 2.

The purpose of material planning is to *match the supply of materials with the demand*. The demand for finished goods is based on external factors such as customer tastes and preferences, economic conditions, and competitors' actions. The demand for the other materials is dependent on the demand for finished goods. Consequently, organizations use different data and processes to plan for different types of materials.

The supply of materials is a function of many internal and external factors. For example, the supply of materials procured externally (e.g., raw materials) depends on availability from vendors as well as the *lead time*, which is the time between placing the order and receiving the shipment. Internally, the supply depends on available production capacity in the factories.

The outcome of material planning is the development of strategic and operational plans that match supply with demand as closely as possible. Excess supply will result in increased inventory costs, which are the expenses associated with storing materials. Insufficient supply will result in a situation called *stock-out* in which the company cannot meet its customers' demands. Both situations can undermine a company's productivity and profits.

Material planning is influenced by the fulfillment process, which provides sales data that companies use to forecast demand for finished goods. It is also influenced by procurement and production, which provide data on lead times and capacities, and by IWM, which provides data on material availability. In turn, material planning will trigger procurement and production processes to ensure that demand is met and IWM processes to ensure that materials are stored until needed.

INVENTORY AND WAREHOUSE MANAGEMENT—STORE

Inventory and warehouse management (IWM) is concerned with the *storage and movement* of materials. For a business to operate efficiently, it is essential that materials be stored so that they can be quickly and easily located when needed. This is especially true for large warehouses where thousands of different materials are stored in large quantities. In addition, companies must be able to move the materials quickly and efficiently to wherever they are needed.

Figure 1-7 depicts four scenarios related to material storage and movement. Quadrant A (top left) shows a request for materials that will be used in the production process. These materials must be located and then issued to the production floor. In Quadrant B, the warehouse receives materials from the production process and then prepares them for storage. This process can include such steps as sorting and determining an appropriate storage location. Finally, the materials are moved into the selected locations. A similar process is used for materials that are received from a vendor via the procurement process (Quadrant C). Finally, when a customer order is processed by the fulfillment process, the warehouse must locate the materials and prepare and send shipments to the customer (Quadrant D).

These examples also clearly illustrate the integration between IWM and procurement, production, and fulfillment. We will discuss integration points with other processes in later chapters.

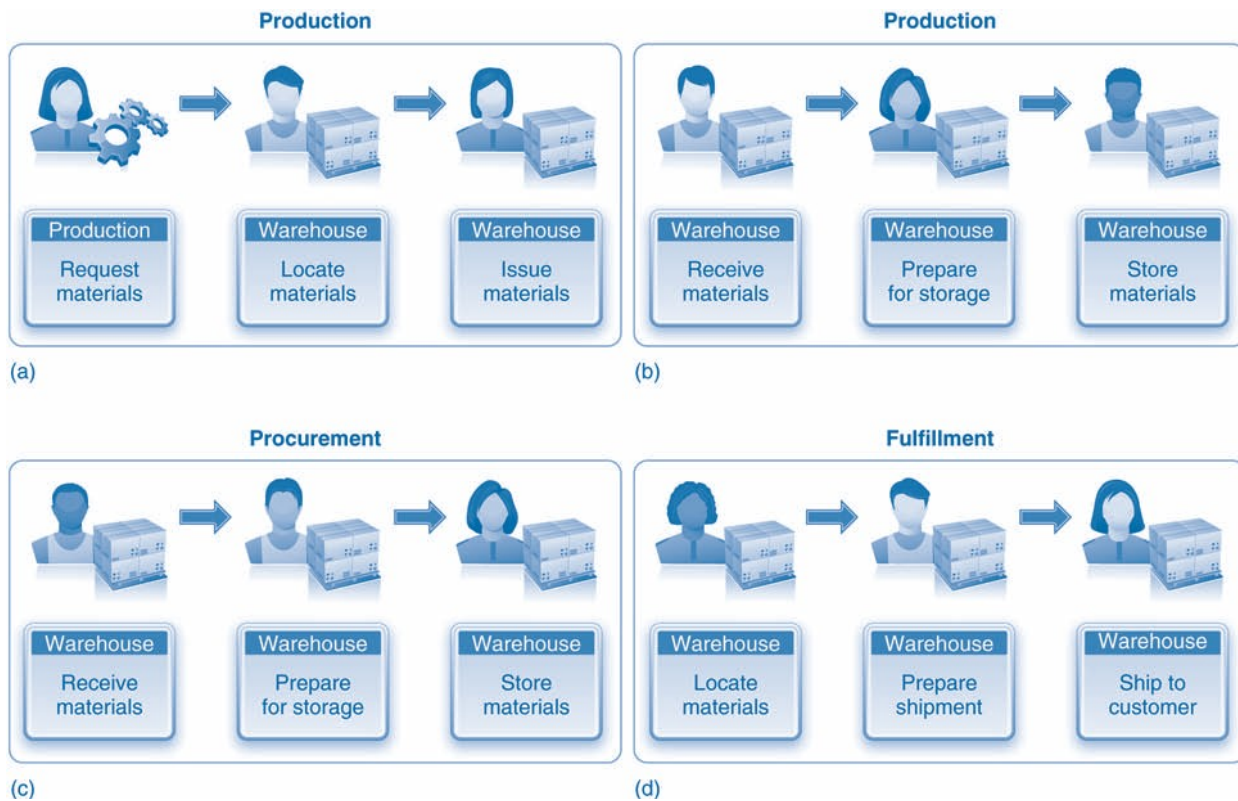


Figure 1-7: Inventory and warehouse management processes

LIFECYCLE DATA MANAGEMENT—DESIGN

A successful organization must constantly improve its products and create new and innovative products that reflect changes in customer tastes and preferences. Lifecycle data management provides a set of tools to *manage product design and improvement* throughout the lifecycle of a product. The product lifecycle begins with idea or concept development; progresses through production, marketing, and service; and concludes when the product is discontinued from the market. It can range from a few months for fad items to many years or even decades for products such as automobiles and bicycles. Products in the latter category typically undergo small but continual improvements over the course of their lifecycle.

Lifecycle data management enables an organization to optimize its product development process, from design to market, while ensuring that it complies with industry, quality, and regulatory standards. At the same time, it provides users—that is, the organization's employees—with access to product data at any point in the product's lifecycle. This capability, in turn, enables the organization to react more quickly to take advantage of market and competitive opportunities.

ASSET MANAGEMENT AND CUSTOMER SERVICE—SERVICE

Asset management is concerned with both the preventive and corrective maintenance of an organization's equipment. *Preventive maintenance* is performed periodically—for example, the routine maintenance of a machine in a factory.

In contrast, *corrective maintenance* is done as needed—for example, repairing a machine when it breaks down. Figure 1-8 illustrates a simplified maintenance process. The trigger is a maintenance request, which can be either preventive or corrective. Production approves the request, and the maintenance is performed. The final stage, settlement, involves an internal charge for the work done.

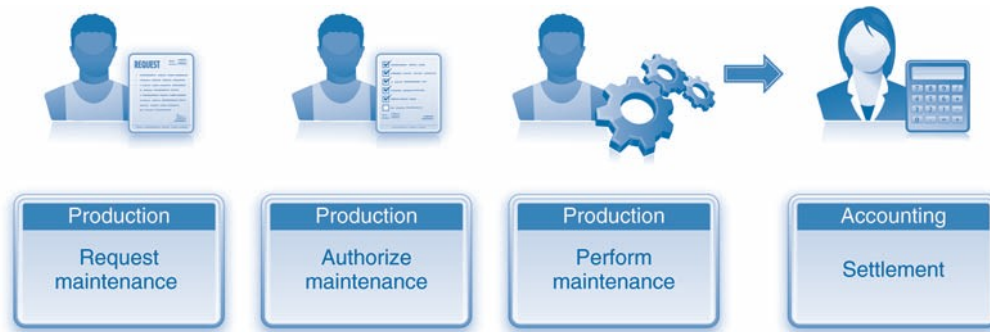


Figure 1-8: An asset management process

A similar process is used for service requests from customers, for example, to *repair a product* they purchased (Figure 1-9). In such cases, different functional areas may be involved. Sales receives a service request, which it approves and forwards to the department responsible for completing the repairs. Settlement will depend on whether the service is covered by a warranty. If it is, then the organization will absorb the cost of the repair. Otherwise, the organization will send an invoice to the customer and then record payment, similar to the steps in the fulfillment process.

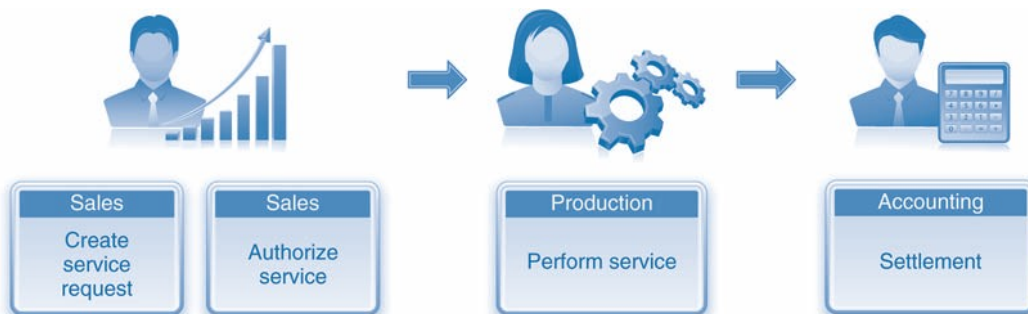


Figure 1-9: A customer service process

HUMAN CAPITAL MANAGEMENT—PEOPLE

Human capital management (HCM) consists of numerous processes related to all aspects of *managing people* in an organization. Examples of HCM processes are *recruitment, hiring, training, compensation and benefits management*, and

payroll administration. In our brief discussion of processes in this chapter, we focused on tasks and the functional areas where they are completed. Clearly, however, it is the people in the functional areas who actually perform the tasks. Consequently, HCM touches every process in the organization. Moreover, it is not uncommon for people in different functional areas to complete many of the tasks in HCM processes. For example, the trigger for recruitment and hiring is a need for people with the requisite skills to complete process tasks. Consequently, the functional area in need of new employees will be involved in this process.

PROJECT MANAGEMENT—PROJECTS

Most business processes are ongoing or repetitive. For example, the lifecycle data management process spans the life of a product, and the procurement and fulfillment processes are repeated frequently. In contrast, a **project** is temporary in nature and is typically associated with large, complex activities, such as the construction of a factory or an aircraft. As we discussed earlier, project management refers to the processes a company uses to *plan and execute large-scale projects*. It involves the use of tools and techniques for managing complex projects.

Projects can be internal or external depending on the recipient of the final outcome. For internal projects, such as constructing a plant, project management is concerned primarily with costs. This is because the outcome of the project benefits the organization and is not sold to a customer. Because no sales are involved, no revenues are created. In contrast, external projects such as building an aircraft for a customer generate both costs and revenues.

Projects rely on resources and capabilities available in other processes. For example, building an aircraft involves purchasing materials (procurement process), making components from these materials (production), supervising people (HCM), and so on. External projects are also integrated with selling to customers (fulfillment). Figure 1-10 illustrates a simplified project management process. The diagram does not identify the specific functional areas in which the work needed to complete the project is performed because this will vary depending on which other processes are involved.

In the planning phase the scope of the project is defined, and the milestones and deadlines are set. The budgeting phase triggers the accounting processes to calculate and allocate the resources needed to execute the project. The project is not executed until management approves the budget. During



Figure 1-10: A project management process

the execution phase the needed processes (e.g., procurement and production) are triggered. In addition, accounting processes are used to keep track of costs and revenues and, for external projects, to issue customer invoices. Finally, throughout the life of the project and at the end of the project an accounting process called *settlement* is periodically carried out to assign costs and revenues to the appropriate parties.

FINANCIAL ACCOUNTING—TRACK FOR EXTERNAL REPORTING

Financial accounting is concerned with *tracking the financial impacts of processes* with the primary goal of meeting legal and regulatory reporting requirements. Thus, it is externally focused. Common reports include the income statement or profit and loss (P&L) statement and the balance sheet. The income statement indicates the organization's financial condition within a specified period of time. It identifies revenues, expenses, and net profit (or loss) for the period. In contrast, a balance sheet indicates the financial condition of an organization at a given point in time. It identifies assets, liabilities, and shareholders' equity. All of these reports must comply with prescribed standards, such as the generally accepted accounting principles (GAAP) in the United States and Handelsgesetzbuch (HGB) in Germany. These reports must be submitted to regulatory agencies at prescribed times, such as annually or quarterly. Finally, these reports are country specific. Therefore, an enterprise that operates in multiple countries must track financial data separately for each country, using that country's prescribed standards.

Various steps in the different processes introduced earlier in this chapter have an impact on an organization's financial status. Organizations analyze this impact using four key processes based in financial accounting: general ledger, accounts receivable, accounts payable, and asset accounting. The *general ledger* process records the impacts of various process steps on a company's financial position. The impacts are recorded in a number of *accounts* in the *general ledger* that represent an organization's income, expenses, assets, and liabilities. These accounts are used to store accounting-relevant data from process steps. *Accounts payable* is associated with the procurement process and is used to track money that is owed to vendors. Similarly, *accounts receivable* is used to track money owed by customers. Accounts receivable and accounts payable automate the general ledger entries associated with the procurement and fulfillment processes so that the financial impact of these processes is recorded automatically. Finally, *asset accounting* is concerned with tracking financial data related to assets such as machinery and cars.

MANAGEMENT ACCOUNTING—TRACK FOR INTERNAL REPORTING

Whereas financial accounting is concerned with external reporting that is mandated by laws and regulations, **management accounting**, or **controlling**, is concerned with *tracking costs and revenues* for internal reporting that is intended to help management control costs and revenues and assess the profitability of various products and market segments. Management creates these reports to support its decision making. Unlike financial accounting reports, management

accounting reports are produced as needed and can contain any information that management deems necessary.

Among the major costs management accounting tracks are materials costs, labor costs, and overhead costs. Management takes these costs into account when it establishes prices for its products or services. It then combines these data with information concerning revenues to determine the profitability of various products and services in different market segments. Ultimately, management utilizes all of this information to make key strategic decisions that affect the organization's products market mix as well as tactical decisions that influence day-to-day operations.

■ GLOBAL BIKE INCORPORATED (GBI)

Throughout this book, we will use the case of **Global Bike Incorporated (GBI)** to illustrate important concepts, processes, and techniques. GBI is a fictional company, and its operations have been greatly simplified to make its business processes and its SAP ERP system easier for you to work with. Although GBI might seem complex as you progress through the chapters of this textbook, rarely will the business operations of a real-world company be as simple as those found in GBI.

Much the data used in this textbook are based on GBI and represent the many aspects of its fictional suppliers, customers, employees, and materials. In addition, all of the hands-on exercises are intended to be completed in a functioning (live) SAP ERP system that is configured with GBI data. (Enterprise Resource Planning, or ERP, systems integrate a company's various functional and cross-functional business processes. We explain ERP systems in greater detail in Chapter 2.) It is very important that you quickly become familiar with GBI and their business operations in order to master the key concepts in this course and to complete your course assignments.

The following paragraphs provide a brief overview of GBI. However, you must retrieve the detailed GBI Annual Report from the *WileyPLUS* course site (explained later) or the *SAP University Alliances Community (SAP UAC)* at <http://uac.sap.com> to learn the full story of how GBI came into existence and to familiarize yourself with the specifics of the company's operations. The SAP UAC is a free site for all university students who are enrolled in an SAP course. The registration process is very simple and is free. In addition to the detailed GBI information on the SAP UAC site, you will also gain access to several SAP career services, including certification information and internship and job opportunities. Your instructor can show you where to access the GBI Annual Report on the SAP UAC site once you have registered.

GBI was founded in 2001 following the merger of two bicycle manufacturers, one based in the United States and the other in Germany. GBI has three lines of business: deluxe and professional touring bikes, men's and women's off-road bikes, and bike accessories. GBI sells its bikes to a network of specialized dealers throughout the world, and it procures its raw materials from a variety of suppliers globally.

GBI has two manufacturing facilities, one in the United States and one in Germany. It also has three additional warehouses, two in the United States and one in Germany. GBI has more than 100 employees globally. The organization uses SAP ERP to support its processes. Figure 1-11 illustrates GBI's enterprise structure.

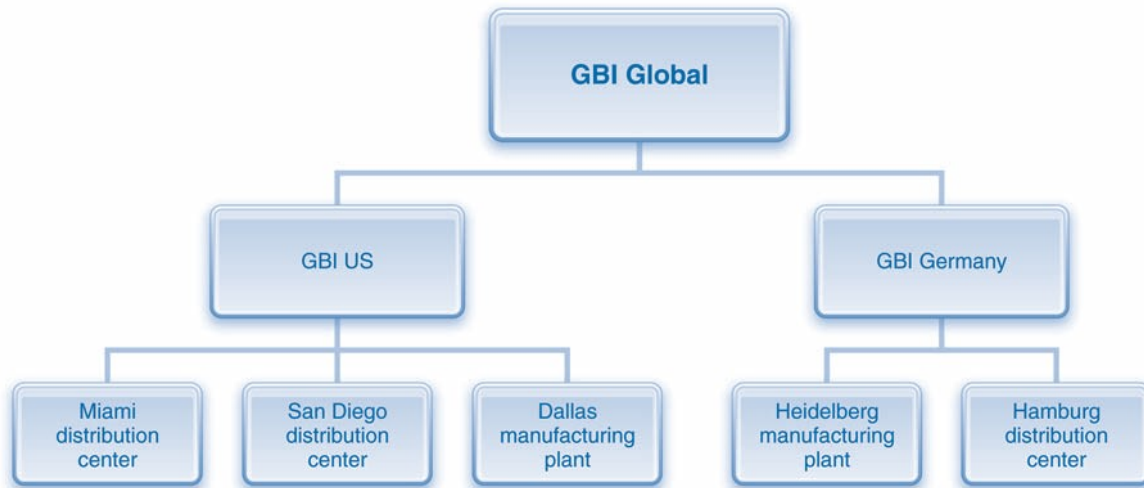


Figure 1-11: GBI enterprise structure

■ HOW TO USE THIS BOOK

This edition of *Integrated Business Processes with ERP Systems* does not cover all of the processes introduced in this chapter. Chapter 2 is an introduction to enterprise systems, and Chapter 3 discusses financial accounting. Chapters 4-8 address the key logistics processes: procurement, fulfillment, production, warehouse management, and material planning. Finally, a concluding chapter discusses all of these processes from an integrated perspective. In addition, the exercises and demonstrations included in this book are available online via *WileyPLUS*.

CHAPTER STRUCTURE

As the previous paragraph indicates, most chapters will focus on a specific process, such as procurement, financial accounting, and production. Most of these chapters share a common structure. Each chapter begins with a review of the simple presentation of the process introduced earlier in this chapter. The chapter then segues to a detailed discussion of data, key concepts, and process steps. General concepts related to data—*organizational data*, *master data*, and *transaction data*—will be introduced in Chapter 2. Each subsequent chapter will focus on the specific concepts and data that are relevant to the process discussed in that chapter. Most chapters require you to understand a few fundamental concepts, which are discussed next. Note that many of the concepts related to organizational and master data are relevant to more than one process. In such cases, we do not repeat the discussion of these concepts. Rather, we refer you to previous chapters where they were discussed.

The third main component of each chapter consists of a detailed discussion of the various steps involved in the process. Each process step will be discussed using the following framework:

- Triggers: the event(s) that cause(s) the process step to be initiated
- Tasks: the specific steps needed to complete the process step

- Data: the information associated with each process step
- Outcomes: the specific products or consequences of the process step

The fourth component of each chapter is reporting. This section examines how the data generated during the process steps are converted into meaningful information. It also provides examples of the types of reports that can help managers improve process performance.

Going further, each chapter employs three pedagogical tools—demonstrations, real-world cases, and exercises—to reinforce the material. Demonstrations illustrate either a particular concept or the execution of a process step in the SAP ERP system. These demonstrations are in the form of animations that can be viewed multiple times. Real-world cases are vignettes from actual companies that illustrate a concept or demonstrate how that company executes a process. Exercises to be completed in an SAP system configured with GBI data provide a final reinforcement of the chapter's key concepts.

SAP SOFTWARE

SAP (pronounced by saying each letter individually, like IBM or ABC) is the pioneer in enterprise systems. SAP was the first company to build a packaged enterprise system, which means that it designed a single piece of software that is used by many companies. Prior to that time, software developers had to create customized software for every company, which was prohibitively expensive.

SAP introduced the first integrated, end-to-end enterprise system, called SAP® R/3, in 1992. The “R” in R/3 stands for “real time.” Prior to the development of enterprise systems, companies typically employed a number of different systems, each of which supported a single function or department. Thus, there were sales systems, accounting systems, manufacturing systems, and so on. These systems were not integrated, so sharing data between and among them was problematic. As you might expect, this architecture regularly experienced delays in executing business processes because data had to be transferred from one system to the next as the process was being performed.

SAP R/3 was designed to eliminate these inefficiencies by executing an entire process from start to finish and consolidating all of the process data in a *single database*. Consequently, regardless of which individuals were completing a step in the process, all of the data were available to them in real time. In addition, everyone else in the company could see the status of the process in real time as well. In today's age of Twitter and RSS feeds, this development might seem trivial. At the time, however, it was a crucial innovation. SAP R/3 was quickly adopted by one major corporation after another, and it catapulted SAP software onto the “must do” list for nearly every large company. By 2010, SAP had more than 110,000 customers in over 120 countries, including nearly every Fortune 1000 company. In 2008, SAP's market share in the ERP category was equivalent to the market share of the next four largest ERP vendors—combined.^{3,4} Today, more than 75% of SAP's customers are small and medium-sized businesses.

³C. Pang, Y. Dharmasthira, C. Eschinger, and K. Motoyoshi, *Market Share: ERP Software, Worldwide*, 2008, July 2008, Gartner.

⁴A. Pang, *Worldwide ERP Applications 2009–2013 Forecast and 2008 Vendor Shares*, October 2008, IDC.

Enterprises of every size, in every industry, all over the world use SAP software to manage their business operations. Regardless of where you live, nearly every major corporation, government entity, and nonprofit organization you are familiar with runs the same SAP software that you will use in this course.

Before you start to think that this book is a marketing brochure for SAP, you should understand why we have explained SAP's strategic importance in business and have selected SAP ERP as the reference system for this textbook. One of the most lucrative and rewarding careers in the IT industry for nearly 20 years has been that of an SAP consultant.⁵ Contrary to what you may have heard, most SAP consultants are *not* programmers. Rather, they are MIS and business majors who have developed a process perspective on business and have become competent in a specific capability of the SAP ERP system. However, even technical programmers who wish to work with SAP must have a deep understanding of how business works in order to program applications that enable business processes to operate more efficiently. In other words, they are people just like you who have mastered the material in this textbook.

Integrated Business Processes with ERP Systems will incorporate a number of demonstrations, examples, and hands-on exercises using SAP ERP. Several other companies offer enterprise systems that have similar capabilities, but it would be very difficult to explain how processes are executed in each of them. We have chosen to include the most prevalent and widely used ERP system that you are likely to encounter in your career. Although some of the concepts in this textbook are specific to SAP ERP, you can easily transfer nearly everything you learn to whichever system is used in the companies where you will work.

When SAP first introduced R/3, almost anybody could claim to be an R/3 expert and thus become a highly paid consultant. Unfortunately, this practice led to quite a few well-publicized project failures. In response, SAP introduced *certifications* for the various modules and technical skills required to be a properly trained consultant. This arrangement enabled consultants who participated in SAP training programs and demonstrated a high degree of skill to distinguish themselves for potential employers. Today, SAP provides more than 100 certification types, classified by solution, focus area, and role. Each certification type specifies three levels of skill: associate, professional, and master. It can take many years and tens of thousands of dollars to progress up to master-level certification. SAP is very proud of the high level of knowledge and skills that are required to earn certification. As you probably suspect by now, the SAP testing process is extremely rigorous. Because an SAP certification is such a highly valued credential, once you have earned one, SAP provides you with a certification number that can be listed on your resume or CV and verified by potential employers—the thousands of consulting companies that implement SAP software and the more than 110,000 (in 2010) companies that run SAP software.

As an added benefit to students enrolled at universities or technical schools that are members of the SAP University Alliances Program, SAP offers special certification academies on campuses around the world where students can earn the same certification as professionals at a reduced cost. This textbook and the additional online materials are based on the content in the

⁵J. Sahadi, Hot 6-figure jobs now, 2007 [Online], *CNN/Money*. http://money.cnn.com/galleries/2007/pf/0708/gallery.hot_six_fig_jobs_now/index.html.

SAP course, which results in an official SAP Associate Application Consultant certification and can be used as a supplement to the SAP course materials. Alternatively, students who master the concepts in this textbook and the additional online materials can take the SAP certification exam at one of over 8,000 global testing centers without participating in a certification academy.

Students who pass the exam will receive the same official SAP certification as working professionals who complete an SAP-sponsored training program. Earning this certification is the first step toward a successful and perhaps lucrative career as an SAP application consultant. Speak with your instructor, and consult the certification information on the SAP University Alliances Community and *WileyPLUS* for more details.

WILEYPLUS

WileyPLUS is an important online supplement to this book. It includes four key components: hands-on exercises, SAP ERP system demonstrations, online chapters, and information about preparing for the SAP certification exam.

The hands-on exercises found on the *WileyPLUS* companion site for this textbook will guide you through the execution of the process steps found in each chapter. You will complete these exercises in a live SAP ERP system based on Global Bike, Inc., which you will become very familiar with throughout the chapters. In essence, you will assume the role of the various employees of GBI and complete their tasks in the SAP ERP system, as if you were doing their job. In this way, you will see the processes executed from multiple viewpoints and thus further develop your “process perspective” of the firm. Your instructor will provide you with login information and the SAP interface to complete these exercises in the SAP ERP system. We anticipate that you will encounter some difficulties while completing the exercises. Nearly every student does, so don’t become discouraged if this happens to you. Your instructor or lab assistant can provide some help if you are working in an on-campus lab. You can also seek help from fellow students in your class. One of the most valuable sources of assistance is the SAP University Alliances (SAP UAC) discussion forum for this textbook. Go to www.extrabandwidth.com/forum, which will redirect you to the SAP UAC forum, where you can ask questions about the text and the exercises. Students and professors around the world will quickly assist you. If you receive a helpful answer, you can award points to the person who helped you. Not only is this a nice form of recognition, but it helps the community reinforce its focus on mutual support and knowledge sharing.

Throughout each chapter you will see references to demonstrations that will illustrate how a particular concept, task, or data is presented in the SAP system. These demonstrations are meant to enhance your conceptual understanding of the topic by providing you with a visual and physical representation of how the system executes or displays a particular component of the process being discussed. We advise you to follow the demonstrations on the *WileyPLUS* website while reading and reviewing the chapter in the printed textbook. For the online version of the textbook on *WileyPLUS*, the demonstrations are linked directly from the references in the chapter for easy access. Going further, the demonstrations are very similar to the activities required in the exercises. Therefore, it will be helpful to review the demonstrations prior to attempting the exercises so as to familiarize yourself with the navigation, key data, and activities required to complete the process steps.

CHAPTER SUMMARY

This chapter introduced the key concepts related to common business processes in organizations, the typical functional structure that companies employ to manage their operations, and the benefits of adopting a holistic view of integrated business processes and their role in effectively translating corporate strategy into operational efficiency.

Most companies are organized according to functional departments, which group together related activities and assets under specialized management controls. Although this approach enables companies to focus resources on specific activities, it also creates communication difficulties and delays between the highly specialized groups. Business processes cut across the vertical barriers (silos) that characterize the functional structure. For this reason they require cross-functional communication and collaborative execution.

Enterprise systems allow companies to effectively manage business processes across functional areas and institutional boundaries. They perform this task by removing barriers to sharing and accessing information, thereby providing a holistic platform to execute integrated business processes consistently and efficiently. One of the key benefits to managing business processes with an integrated enterprise system is that process data are collected throughout the execution of each step of the process. Managers can then use these data to monitor and improve the organization's processes. Enterprise systems enable companies to achieve operational efficiency through transparency across functional areas, and they provide consistent information for managerial decision making. All business processes have an impact on the organization's financial status, and the real-time impact of process execution can be monitored and analyzed through the use of an integrated enterprise system.

KEY TERMS

Asset management and customer service processes	Inventory and warehouse management (IWM) process
Business process	Lifecycle data management process
Controlling	Management accounting or controlling (CO) processes
Enterprise systems (ES)	Material
Financial accounting (FI) processes	Material planning process
Fulfillment process	Procurement process
Functional structure	Production process
Global Bike Incorporated (GBI)	Project
Human capital management (HCM) processes	Project management processes
	Silo effect

REVIEW QUESTIONS

1. Describe the functional organizational structure. Why do you think this structure is so widely used?
 2. What is the silo effect? Why does it exist? What problems does it create? How can an organization reduce or eliminate the silo effect?
 3. What is a business process? Why is adopting a process view of organizations essential to becoming a successful manager?
 4. Briefly describe the key business processes included in this chapter in terms of their key steps.
 5. Explain the interrelationships among the key processes included in this chapter. Why are these interrelationships important?
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Introduction to Enterprise Systems

LEARNING OBJECTIVES

After completing this chapter you will be able to:

1. Discuss the evolution and key business benefits of enterprise systems.
2. Explain the role of enterprise systems in supporting business processes.
3. Differentiate the different categories of data within SAP® ERP.
4. Identify and analyze the major options for reporting.

In Chapter 1 we discussed the key processes executed in organizations. We saw that there is a high degree of integration and interdependence among processes. Even within small organizations it is difficult to execute these processes efficiently using manual techniques. Thus, organizations invariably rely on the use of enterprise systems to ensure that these processes are executed efficiently and effectively. In this chapter we discuss key concepts in enterprise systems, particularly SAP ERP. We begin by expanding on the introduction to enterprise systems and SAP ERP provided in Chapter 1. We follow this discussion with a brief history and evolution of SAP ERP. We then explore the different types of data utilized in SAP ERP. Finally, we conclude the chapter by examining various reporting options enabled by enterprise systems (ES).

■ ENTERPRISE SYSTEMS

Enterprise systems are one of the most complex and powerful information systems in use today. As we explained in Chapter 1, this book will cover the world's most popular ES—SAP ERP—in great depth. Although you might have studied enterprise systems in a previous course, we provide a quick review of the key concepts to refresh your knowledge and to place these concepts in an appropriate context for the rest of this course. In this section we briefly discuss the architecture of enterprise systems, the SAP® Business Suite, SAP ERP, and the technology platform—SAP® Netweaver—that forms the foundation for these applications.

ARCHITECTURE OF ENTERPRISE SYSTEMS

The **architecture** of an enterprise system refers to the technical structure of the software, the ways that users interact with the software, and the ways the software is physically managed on computer hardware. Most modern ES have either a three-tier client-server architecture or a service-oriented architecture. There are many different ways to deploy ES in these two architectures. Both models offer distinctive technical and cost benefits, and both models have drawbacks. Nevertheless, the impact of these two models on the management of business processes is largely the same. We examine both types of architecture below.

Client-Server Architecture

Think of a desktop application that you routinely use, such as word processing, spreadsheet, or presentation software. These applications consist of three components, or *layers*: (1) how you interact with the application (using menus, typing, and selecting); (2) what the application allows you to do (create formulas or charts, compose an essay); and (3) where the application stores your work (on your hard drive or flash drive). These layers are the **presentation layer**, **application layer**, and **data layer**, respectively. In the desktop applications mentioned above, all three layers are contained in one system. In contrast, the **three-tier client-server architecture** separates these layers into three separate systems, as illustrated in Figure 2-1.

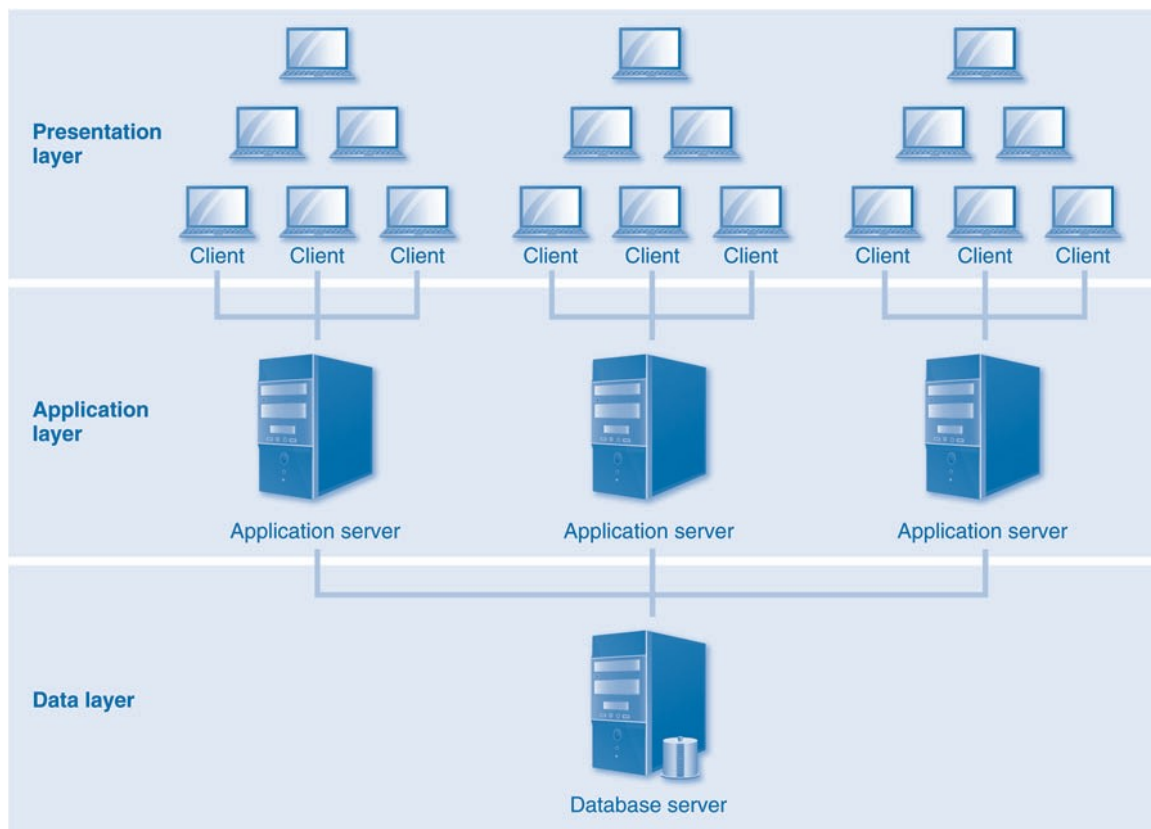


Figure 2-1: Three layers of the client-server architecture

Much of the work you do on the Internet uses a three-tier architecture. Your browser is the presentation layer. Through your browser, you connect to many systems (websites) that provide a variety of capabilities (e-mail, purchasing goods, information sharing). These websites contain the applications that execute the request you send through the browser (via HTTP), and they retrieve and store data in a connected database.

The shift to the three-tier client-server architecture dramatically reduced the costs of acquiring, implementing, and using an ES while significantly increasing the scalability of the systems. *Scalability* refers to the ability of the hardware and software to support a greater number of users easily over time, typically at a decreasing cost per user. These two benefits transformed ES from a capability that only a few large companies could afford into a technology that tens of thousands of companies now utilize.

Service-Oriented Architecture

In the early 2000s, companies began to Web-enable their three-tier applications so that users could access the systems through a Web browser. During these years companies also benefited from new technologies that could help link, or *integrate*, many different client-server systems together in new and valuable ways. These new technologies are collectively labeled **service-oriented architecture**, or SOA. The fundamental concept behind SOA relates to the technical capabilities that allow systems to connect with one another through standardized interfaces called *Web services*. By using Web services, companies could now integrate multiple client-server applications and create enterprise *mash-ups*, or *composite applications*. Composite applications and mash-ups rely on Web services to send and receive data between and among ES in a standardized way, which eliminates a great deal of cost and complexity from integration projects. In addition, they execute newer and more specific processes than are typically found in the standard ES.

Companies such as SAP have invested billions of dollars to service-enable their applications so that these systems can be exposed—that is, their functionality can be made visible to users—and can be connected to a great number of composite applications. By using SOA to integrate and expose the business processes and data inside an ES, companies can now create new composite applications quickly and inexpensively. In essence, SOA enables companies to build composite applications on top of their existing three-tier client-server applications without changing the underlying applications. This capability gives companies an entirely new level of flexibility at an extremely low cost.

ENTERPRISE RESOURCE PLANNING (ERP) SYSTEMS

Enterprise resource planning (ERP) systems are the world's largest and most complex ES. ERP systems focus primarily on *intra-company processes*—that is, the operations that are performed within an organization—and they integrate functional and cross-functional business processes. Typical ERP systems support Operations (Production), Human Resources, Finance & Accounting, Sales & Distribution, and Procurement. As we discussed in Chapter 1, SAP was the first company to create a fully integrated and global ERP system, SAP® R/3, which could manage end-to-end processes for companies that operated in many different countries, with multiple languages and



Figure 2-2: The SAP ERP solution map. Copyright SAP AG 2011

currencies. Figure 2-2 shows the solution map for the current version of the system developed by SAP, known as SAP ERP.

The solution map identifies the functionality and processes supported by the system. Notice that many of the functional capabilities in the solution map are similar to the business processes that were defined in Chapter 1. In addition, several areas of SAP ERP overlap with functional groups within the company. These overlaps are a result of the tight integration of ERP and the processes that it manages. As companies have adopted more and more ERP capabilities and begun to view their companies from more of a process perspective, the ES world and the functional world have begun to merge. Given the scope and size of the SAP ERP system, we will focus on the core ERP functional modules in this book.

Although companies are moving toward a process view of organizations, and our book takes a process view of business operations, the functional view still persists in many organizations. The capabilities of an ERP system are often described in terms of modules or specific capabilities, and it is still quite common to see or hear SAP ERP referred to in terms of module abbreviations in job advertisements or industry discussions. Figure 2-3 lists the more common

- | | |
|-------------------------------|--|
| • Production Planning (PP) | • Financial Accounting (FI) |
| • Materials Management (MM) | • Management Accounting / Controlling (CO) |
| • Sales and Distribution (SD) | • Human Resources (HR) |
| • Plant Maintenance (PM) | • Business Intelligence (BI) |
| • Project Systems (PS) | |
| • Quality Management (QM) | |

Figure 2-3: SAP ERP modules

modules in SAP ERP and the abbreviations that are typically used for them. For example, a person with expertise in the financial accounting and management accounting modules of SAP ERP is typically called a FICO expert based on the abbreviations of the two modules.

As more companies acquired ERP systems, the next step in the evolution of ES was to connect these systems so they could support *inter-company processes*—that is, processes that take place between and among companies. Examples of inter-company systems are **supply chain management (SCM)** and **supplier relationship management (SRM)** systems, which connect a company's ERP system to those of its suppliers. SCM connects a company to other companies that supply the materials it needs to make its products. Typical SCM systems help companies plan for their production requirements and optimize complex transportation and logistics for materials. SRM systems typically manage the overall relationships with the materials suppliers. SRM systems contain functionality to manage the quotation and contracts processes. These systems act as extensions to the procurement and material planning processes of ERP systems.

On the other side of the manufacturing and sales processes, **customer relationship management (CRM)** systems connect a company's ERP system to those of its customers. CRM systems provide companies with capabilities to manage marketing, sales, and customer service. These systems are an extension of the fulfillment process of ERP systems. **Product lifecycle management (PLM)** systems help companies administer the processes of research, design, and product management. In effect, PLM systems help companies take new product ideas from the virtual drawing board all the way to the manufacturing facility.

The collection of these inter-company systems and the underlying intra-company ERP system is called an **application suite**. Suite vendors, such as SAP and Oracle, provide fairly comprehensive collections of applications that offer an enormous amount of functionality and cover most of the standard business processes.

Figure 2-4 identifies the various capabilities that are part of an application suite and illustrates how they connect to other members of a company's business network. It is important to note that one of the key benefits of utilizing a complete suite of software is that the data and processes are integrated among the systems in the suite. That is, although they are separate systems, they are designed so that they work together in an integrated manner.

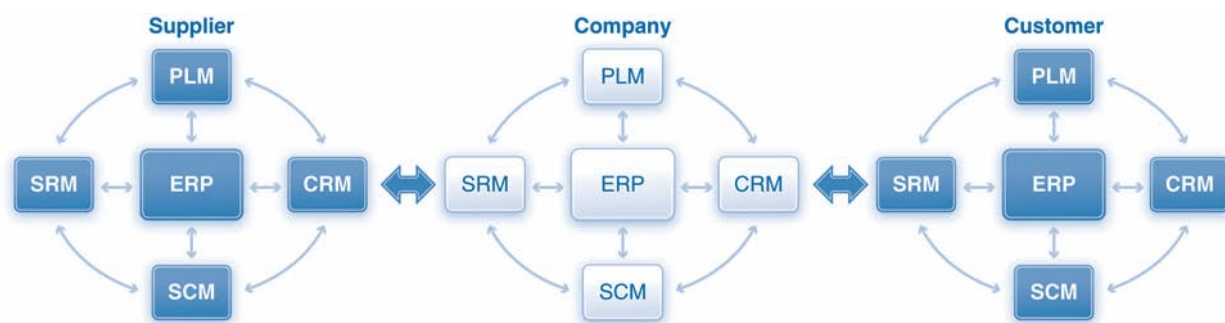


Figure 2-4: The ES application suite

The focus of this textbook is on core intra-company processes and ERP systems. Keep in mind, however, that the emergence of inter-company business capabilities is one of the most important developments in the modern business environment. A fundamental understanding of the key business processes and ERP systems is a prerequisite to studying advanced topics such as supply chain management and customer relationship management because those processes are extensions of the core ERP-enabled business processes.

APPLICATION PLATFORMS

Another critical component of ES is application platforms. Much like the role of the operating system for your personal computer, **application platforms** serve as a type of “enterprise operating system” for a company’s ES landscape by allowing all of the various systems to communicate seamlessly with one another as well as with systems outside the company. SAP introduced its application platform, SAP NetWeaver, in 2003. SAP NetWeaver is now an integral part of SAP ERP and the SAP Business Suite. It contains the SOA capabilities needed to integrate SAP systems with non-SAP systems. In addition, it provides companies with a toolset to build new composite applications or to plug in independent software vendor (ISV) applications on top of their core ERP and suite applications. Figure 2-5 illustrates how SAP NetWeaver interacts with the SAP Business Suite of applications.

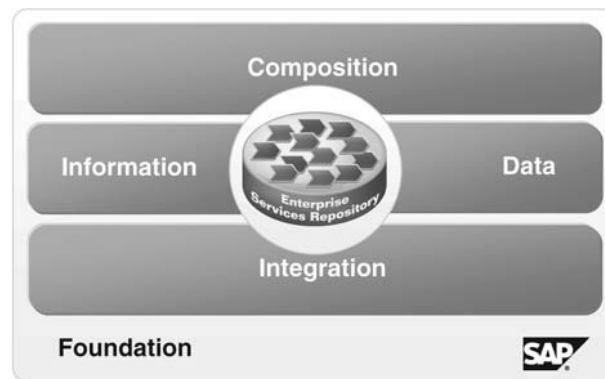


Figure 2-5: SAP NetWeaver. Copyright SAP AG 2011

The SAP Business Suite, which includes SAP ERP, SAP® CRM, SAP® SCM, SAP® PLM, and SAP® SRM, runs on SAP NetWeaver. You can think of the relationship between the SAP Business Suite and SAP NetWeaver in terms of the relationship between Microsoft Office and the Microsoft Windows operating system. The SAP Business Suite consists of the applications that companies use to manage and execute their business processes, just as Microsoft Office contains the applications that you use to accomplish tasks such as creating a presentation, a document, or a spreadsheet. SAP NetWeaver orchestrates the communication among the applications, transporting the master data and technical information needed by the applications at different points in the process. Similarly, Microsoft Windows is an operating system that the Microsoft Office applications use to communicate with one another and to access other

capabilities and tools. SAP NetWeaver also contains several technical tools to help companies extend the SAP Business Suite applications, integrate with other non-SAP applications, and build (compose) new applications. In essence, SAP NetWeaver is the “operating system” for a company’s entire range of business processes.

■ DATA IN AN ENTERPRISE SYSTEM

As we discussed earlier, a central component of any ERP system is the common database that stores data related to all the processes. Without this function, integrating the various processes would be difficult, if not impossible. Therefore it is essential to understand how data are organized in an ERP system. We address this topic in the following section. We then introduce the different types of data that are stored in an ERP system, and we identify basic data elements that are common to many processes. We will develop these topics and introduce additional data elements in later chapters that discuss specific processes. For the purposes of this chapter we will restrict our discussions to the procurement and fulfillment processes introduced in Chapter 1.

Data in an ERP system are used to represent the physical system in which process steps such as creating a purchase order and receiving goods are carried out. These steps generate data, which represent the outcomes of the steps. There are three types of data in an ERP system: organizational data, master data, and transaction data.

ORGANIZATIONAL DATA

Organizational data are used to represent the structure of an enterprise. Examples of organizational structure are companies, subsidiaries, factories, warehouses, storage areas, and sales regions. The three organizational data elements discussed in this chapter are client, company code, and plant (see Figure 2-6). Note that the terms *organizational data*, *organizational levels*, and

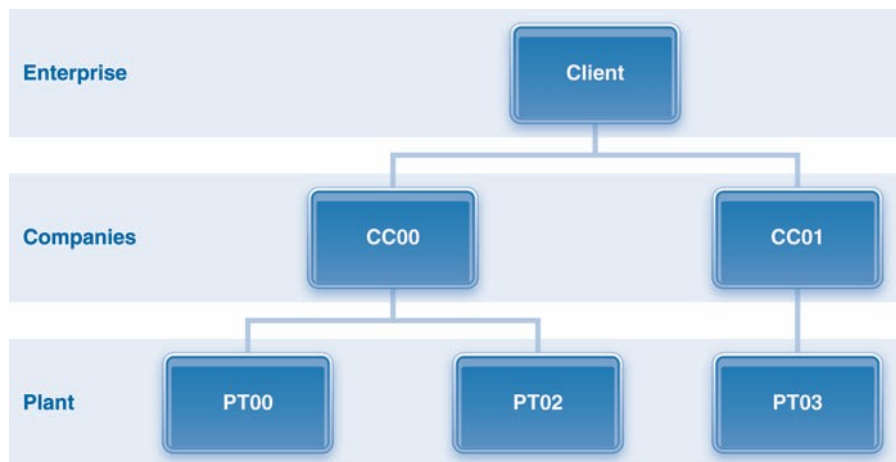


Figure 2-6: Organizational data

organizational elements are often used interchangeably, depending on the context. These data are relevant to many of the processes discussed in Chapter 1. We will introduce additional organizational data as needed in the various process chapters.

Client and Company Code

A **client** is the highest organizational level in SAP ERP. It represents an enterprise consisting of many companies or subsidiaries. Each company within the enterprise is represented by a **company code**. Each company code represents a separate legal entity, and it is the central organizational element in financial accounting. That is, financial statements required for legal reporting purposes are maintained at the company code level. A client can have multiple company codes, but a company code must belong to only one client.

Figure 2-7 shows organizational data for GBI. Recall from Chapter 1 that GBI consists of two companies, one in the United States and one in Germany. GBI is represented by a client, and each of the two companies is represented by a company code, US00 and DE00, respectively.

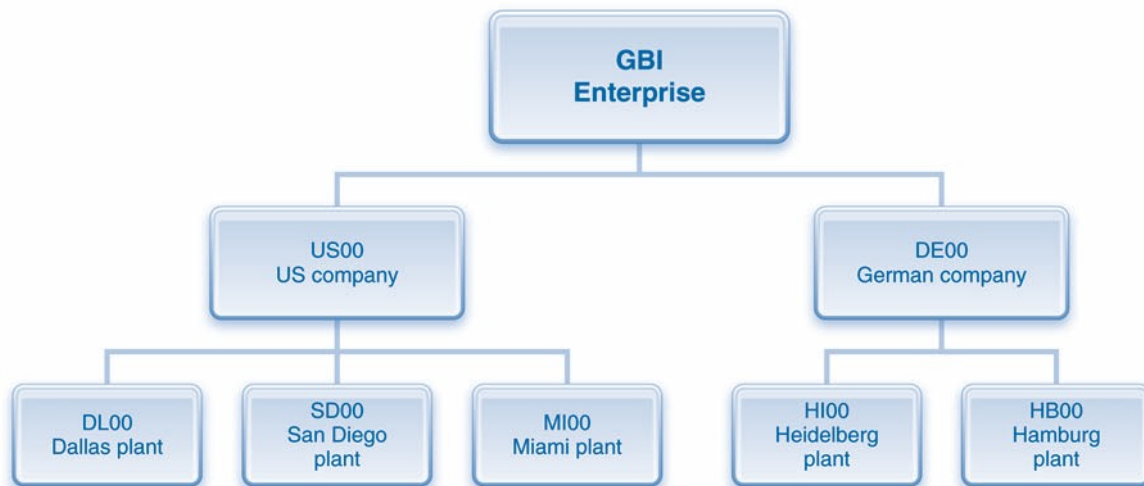


Figure 2-7: GBI organizational data

Plant

A **plant** is an organizational element that performs multiple functions and is relevant to several processes. It is essentially a facility in which the following functions are performed:

- Products and services are created.
- Materials are stored and used for distribution.

- Production planning is carried out.
- Service or maintenance is performed.

A plant can be a factory, a warehouse, a regional distribution center, a service center, or an office. It can be a part of a building, an entire building, or a collection of buildings. In addition, a single building can house multiple plants. Consider, for example, a company that has several offices in a building. Different services or activities, such as processing customer returns and providing technical support to customers, are performed in these offices. Each of these offices is defined as a separate plant. Just as a client can have multiple company codes, a company code can contain multiple plants. However, a plant can belong to only one company code.

As illustrated in Figure 2-7, GBI operates five plants for the manufacture and storage of bicycles and accessories. Three plants are located in the United States—in Dallas (DL00), San Diego (SD00), and Miami (MI00). The Dallas plant is a manufacturing facility, whereas the other two are distribution centers from which products are shipped to customers. The Dallas plant ships products to customers as well. The other two plants are located in Germany—in Hamburg (HB00) and Heidelberg (HI00). The Heidelberg plant operates as both a manufacturing facility and a distribution center, while the Hamburg plant is exclusively a distribution center.

Business Processes in Practice 2.1: Coca-Cola® Enterprises

Coca-Cola Enterprises provides us with a good example of a company that has several types of plants. Coca-Cola operates factories that produce both raw materials and finished goods. Some factories produce the syrup that forms the basis of the Coca-Cola products you are familiar with. Other factories combine the syrup with carbonated water during the bottling process to make the finished Coca-Cola products. The finished goods, which consist of cases of bottles and cans of Coca-Cola

products, are then shipped to regional distribution centers for storage until they are transported to end customers, such as retailers. The syrup factory, the bottling factory, and the distribution center are all considered plants in Coca-Cola's SAP ERP system.

Source: Coca-Cola Company Reports & CCE CIO presentation at SAP Sapphire 2008.

MASTER DATA

Master data represent entities associated with various processes. For example, processes involve buying materials from vendors and selling materials to customers. In this example, customers, vendors, and materials are represented in an ERP system using master data.

The most commonly used master data in an organization is the material master. Materials are used in numerous processes. They are purchased, sold, produced, and planned for. They are used in maintenance and service, and in projects. Consequently, material master data are some of the most complex and extensively utilized data in an ERP system. In contrast, other master

data are relevant only to certain processes. For example, vendor master data apply to procurement, and customer master data are utilized in fulfillment. This chapter will focus exclusively on material master data. We will consider other master data in the appropriate process chapters.

Material Master

Although materials are relevant to many processes, each process uses the materials differently. For example, the procurement process buys materials, the production process makes materials, and the fulfillment process sells materials. Each process, therefore, requires data about the material that may or may not be needed by other processes. For example, the procurement process requires data concerning who is responsible for purchasing the material and how much should be ordered. Similarly, the fulfillment process utilizes data concerning product availability and shipping conditions.

It should be evident that because the material master is used in numerous processes, it must include a large amount of data. To manage these data, the material master groups them into different categories or *views*, each of which is relevant to one or more processes. Figure 2-8 illustrates the major views of data in the material master. Basic data, such as *material number*, *description*, and *weight*, are relevant to almost all processes. The other views are relevant only to certain processes. For example, purchasing data are relevant to the procurement process and sales data to the fulfillment process. The data in each of the views will be explained in detail in the appropriate process chapters.

Material Types

Another factor that influences the type of data or view needed is the type of material. Materials are categorized into different **material types** based on the



Figure 2-8: Material master data

way they are used in the firm's operations. Each material type has different characteristics and is used for different purposes and in different ways. The material type determines which business processes are permitted to use the material. This, in turn, determines which types of data must be maintained for the material. Therefore, the data in the material master will vary for different types of materials. Specifically, material type determines, for example, the screens that appear in the material master record, the department or function-specific data that must be maintained, how the material numbers are determined, the appropriate procurement types (in-house or external), and the general ledger accounts that need to be updated. Table 2-1 provides a list of commonly used material types.

Material Type	Description
DIEN	Services
ERSA	Spare parts
FERT	Finished goods
FHMI	Production resources/tools
HALB	Semifinished goods
HAWA	Trading goods
HIBE	Operating supplies
IBAU	Maintenance assembly
KMAT	Configurable material
LEER	Empty containers
NLAG	Nonstock, nonvaluated material
PROD	Product group
ROH	Raw materials
UNBW	Nonvaluated, stocked material
VERP	Customer returnable packaging
WETT	Competitive products

Table 2-1: SAP material types

The four most common material types are raw materials, semifinished goods, finished goods, and trading goods. We describe and discuss these material types below. The terms in parentheses following the names are the SAP ERP abbreviation for that material type. These abbreviations are based on the German term for each type. GBI's product structure (finished goods and trading goods) is presented in Figure 2-9. Table 2-2 lists GBI materials grouped by material type.

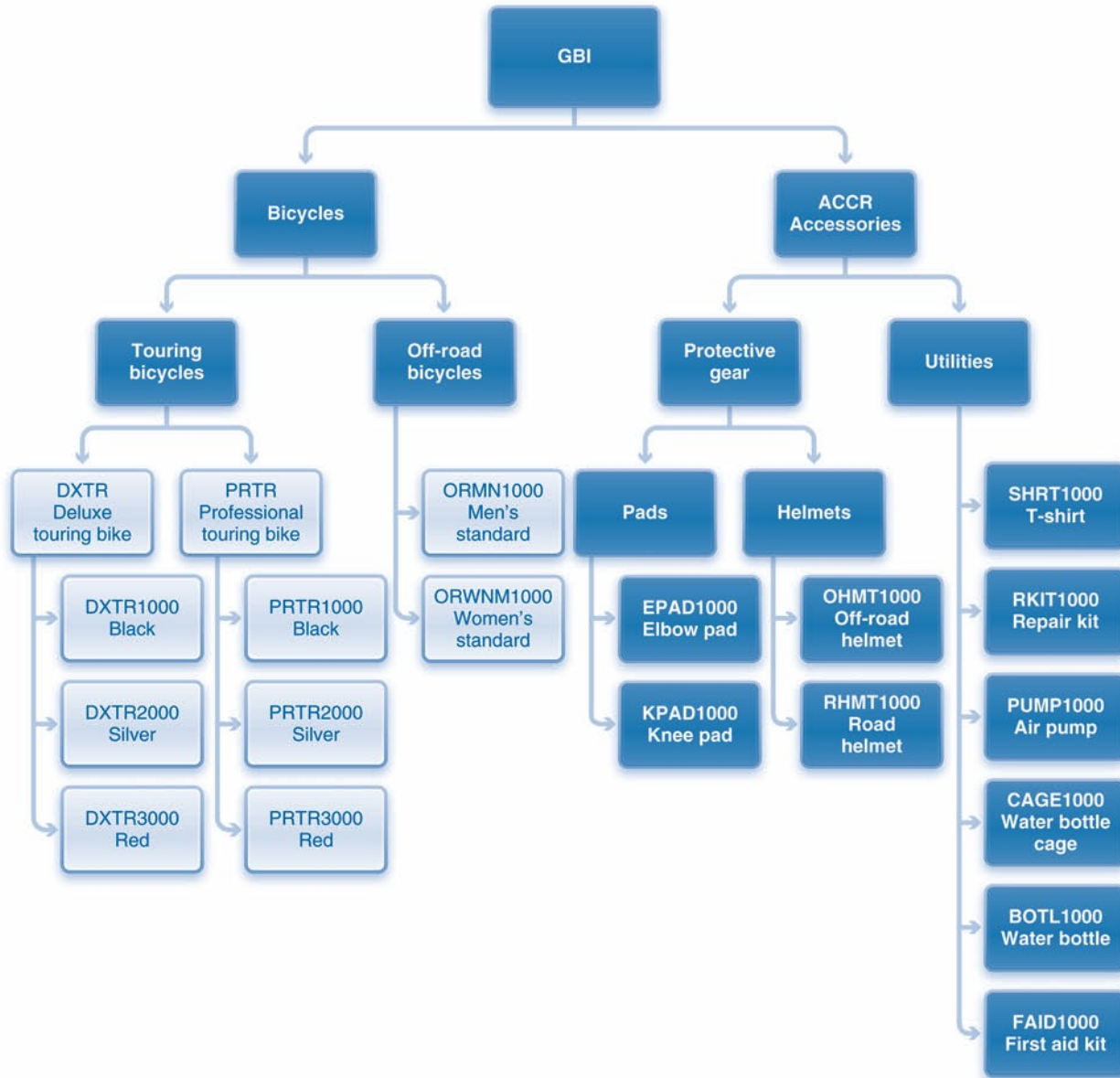


Figure 2-9: GBI product structure

- **Raw materials** (ROH) are purchased from an external source—a vendor—and used in the production process. Typically, raw materials are not sold to end-customers. Consequently, the material master will contain data related to procurement and production but not fulfillment. Examples of raw materials utilized by GBI are frames, wheels, tires, and tubes.
- **Semifinished goods** (HALB) are typically produced in-house from other materials (e.g., raw materials) and are used in the production of a finished good. Consequently, data related to production must be maintained for semifinished goods. Front wheel assemblies are

Raw Materials	Semifinished Goods	Finished Goods	Trading Goods
• Deraillleur gear assembly • Frame • Seat kit • Handle bar • Pedal assembly • Chain • Brake kit • Warranty document • Packaging • Tire • Tube • Wheel • Nuts and bolts	• Wheel assembly	• Deluxe touring bike (3 colors) • Professional touring bike (3 colors) • Men's standard off-road bike • Women's standard off-road bike	• Elbow ads • Knee pads • Off-road helmet • Road helmets • Repair kit • Air pump • Water bottle cage • First aid kit • T-shirt

Table 2–2: GBI material list

an example of semifinished goods from GBI. GBI purchases tires, wheels, and tubes and then uses these raw materials to create wheel assemblies.

- **Finished goods** (FERT) are created by the production process from other materials, such as raw materials and semifinished goods. They are generally not purchased. As a result, the material master for finished goods will include data related to production and fulfillment, but not procurement. An example of a finished good from GBI is the deluxe touring bicycle, which is produced from raw materials (e.g., frames) and semifinished goods (e.g., wheel assemblies).
- **Trading goods** (HAWA), like raw materials, are purchased from a vendor. Unlike raw materials, however, trading goods are resold to customers. Significantly, the company does not perform any additional processing of the material prior to reselling it. Therefore, the material master for trading goods will include data related to purchasing and selling but not production. An example of a trading good from GBI is a helmet. GBI simply purchases the helmets from a supplier and resells them to its customers; no other steps are involved.

Material Groups

Related to material type is the concept of a **material group**, which includes materials with similar characteristics. For example, all of the materials used in the production of bicycles, such as tires and tubes—which are raw materials—and wheel assemblies—which are semifinished goods—can be included in one material group called *production*. As another example, all bicycles, which are

generally finished goods but can also be trading goods, can be placed into one group called *sales*. Alternatively, bicycles can be grouped based on how they are used, such as *touring* and *off-road*. In the retail industry, material groups represent merchandise categories such as footwear, clothes, and beverages. Materials are grouped so that they can be managed collectively. For example, planning for off-road bicycles is performed for all of the bikes in that material group rather than for individual bikes or brand names.

Organizational Level

A final factor that determines the type of data included in the material master is organizational level. Materials can be defined differently for different organizational levels. For example, the same material can be used in multiple plants, but the ways it is used can vary from one plant to another. If a company exports materials from only one of its plants, for instance, then data related to exporting that material must be included in the material's definition for that plant. These data are not necessary, however, for the other plants. As another example, GBI may choose not to ship bikes to customers from its Dallas plant. Rather, it sends the bikes to its two distribution centers (Miami or San Diego), which then ship them to customers. In this case, sales-related data for the bikes are included in the material master's definition for the Miami and San Diego plants but not for the Dallas plant.

Demo 2.1: Review material types

Demo 2.2: Review material master data

TRANSACTION DATA

Processes are executed in the context of organizational levels, involve master data, and result in transaction data. **Transaction data** reflect the consequences of executing process steps, or *transactions*. Examples of transaction data are dates, quantities, prices, and payment and delivery terms. Thus, transaction data are a combination of organizational data, master data, and situational data—that is, data that are specific to the task being executed, such as who, what, when, and where. The composition of transaction data is illustrated in Figure 2-10.

SAP ERP uses several different types of documents to record transaction data. Some of these documents are created or utilized as the process is being executed; others record data after the process steps are completed. We refer to the first category as **transaction documents**. Examples are purchase orders, packing lists, and invoices. A purchase order communicates the company's order to its vendor. A packing list accompanies the shipment sent by the vendor, and an invoice is a request for payment for materials shipped.

Documents that record data generated after the process steps have been completed include **financial accounting [FI] documents, management**

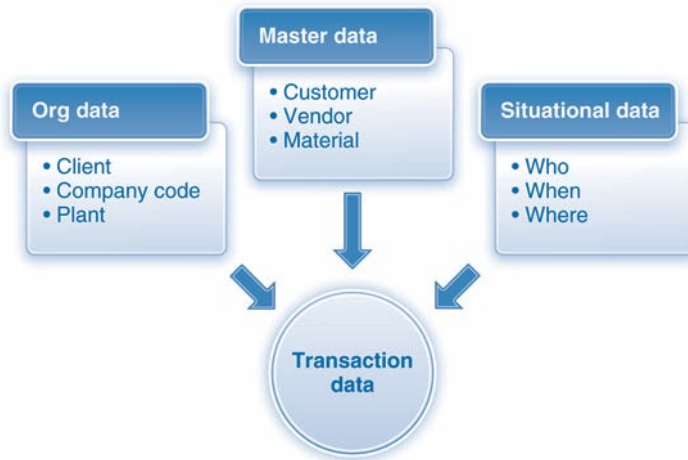


Figure 2-10: Transaction data

accounting or controlling [CO] documents, and **material documents**. These three documents are “virtual” documents in that they reside in the enterprise system and are printed only occasionally as needed. FI and CO documents record the financial impact of process steps. For example, when a company receives a payment from a customer, there is a financial impact, and an FI document is created. Material documents record materials movements, such as when materials are received from a vendor or shipped to a customer.

Documents typically consist of two sections, a *header* section and a *detail* or *line item* section. Figure 2-11 illustrates the concept of headers and items using an example of a purchase order, which is a type of transaction document. The top part of the document is the header. The three materials included in the purchase order—knee pads, elbow pads, and off-road helmets—are the line items. The header includes data such as purchase order number, date, and payment terms that are relevant to all line items, meaning that these data are relevant to the entire document. In contrast, the data in the line items, such as quantity and price, are specific to each item. A document can include multiple line items, as illustrated in the figure.

Demo 2.3: Review a purchase order

REPORTING

As you have probably concluded from the discussions of master data, organizational data, and transaction data, enterprise systems produce and consume massive amounts of data in the day-to-day execution of business processes. In fact, it is not uncommon for companies to have several terabytes (trillions of bytes) of “live” data passing through their ERP systems on a weekly basis. More importantly, after data are no longer needed for process execution, they

Global Bicycle Incorporated 5215 N. O'Conner Blvd. Dallas, Texas, 75039 Phone: +1.972.555.2000 Fax: +1.972.555.2001		PURCHASE ORDER Purchase Order Number: 4546			
THE PURCHASE ORDER NUMBER MUST APPEAR ON ALL RELATED CORRESPONDENCE, SHIPPING PAPERS, AND INVOICES					
TO: Olympic Protective Gear 2100 Summit Boulevard Atlanta, GA, 30319		SHIP TO: GBI San Diego Distribution Center 150 Spear Street San Diego, 94105 +1.415.555.7700			
Purchase Order #	P.O. Date	Delivery Date	Shipped VIA	F.O.B. Point	Payment Terms
4546	July 11, 2009	July 27, 2009	Ground	Destination	Net 30
Quantity	Material #	Material Description	Unit Type	Unit Price	Item Total
100	KPAD1000	Knee Pads	Each	37.50	3,750.00
100	EPAD1000	Elbow Pads	Each	37.50	3,750.00
50	OHMT1000	Off-road Helmets	Each	25.00	1,250.00
SUBTOTAL					\$8,750.00
SALES TAX					Exempt
SHIPPING AND HANDLING					Included
OTHER					N/A
ORDER TOTAL					\$8,750.00
Authorized by: _____ Purchasing Manager			Date: _____		

Figure 2-11: Purchase order

then become historical data and must be archived securely and made available for many types of data analysis. This process generates many more terabytes of historical data that are stored in complex data repositories called *data warehouses*. With such a massive volume of data to deal with, how can a company extract the meaningful information it needs to make better decisions and operate more efficiently?

At the most basic level, every enterprise system contains transaction and historical data in its main database. Transaction data relate to processes that are currently in use or have been completed recently, within days or weeks.

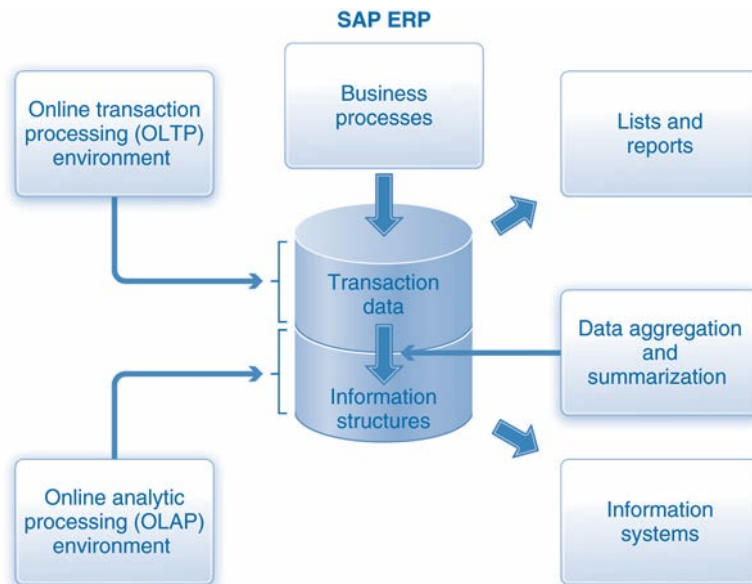


Figure 2-12: Reporting options within SAP ERP

Conversely, historical data are typically comprised of transaction data for processes that have been completed within months or years. **Reporting** is a general term used to describe the ways that users can view and analyze both transaction and historical data to help them make decisions and complete their tasks. Reporting capabilities range from simple lists of information for basic users to analytical tools that can perform powerful statistical analysis and advanced calculations to produce extremely detailed information for specialist users. The data used in the different types of reporting are the same, but the ways in which they are extracted, filtered, and presented differ greatly and involve varying levels of complexity and expertise, depending on how employees plan to use them.

SAP ERP provides two reporting options—simple *lists* of data and documents and *analytics*. Before we examine these options, however, we need to distinguish between the *transactional environment* and the *analytic environment* of SAP ERP (see Figure 2-12). The transactional environment of SAP ERP is an **online transaction processing (OLTP)** system, which, as the name suggests, is designed to capture and store detailed transaction data. The primary function of OLTP is to execute process steps quickly and efficiently; it is optimized for this purpose. OLTP is not used to generate sophisticated reports because it lacks the computing power to parse through and analyze the vast stores of data that most companies accumulate. Consequently, businesses employ OLTP to generate only simple lists and reports.

For detailed data analysis, SAP ERP includes an **online analytic processing (OLAP)** environment in the form of *information systems*. Instead of using detailed transaction data, these systems use information structures to provide analytic capabilities. **Information structures** capture and store specified transaction data in an aggregated and summarized form that enables users to analyze the data as needed. Each information structure in the OLAP environment is defined in terms of three features: characteristics, key figures, and period definition (Figure 2-13). **Characteristics** are the objects for which

Period	Characteristic		Key figures	
Date	Customer	Material	Sales quantity	Sales amount
5/12/09	Rocky Mountain Bikes	DXTR1000	23	\$64,400
5/19/09	Philly Bikes	PRTR1000	45	\$135,000
5/23/09	Beantown Bikes	DXTR1000	34	\$95,200
....				

Figure 2-13: Components of information structures

data are collected. These objects are typically organizational data such as plant and sales organization and master data such as materials, vendors, and customers. An information structure can include up to nine characteristics. **Key figures** are performance measures, such as quantities and counts that are associated with the characteristics. Examples are number of orders, quantities ordered, order value, and invoice amounts. Thus, instead of recording the details of orders and invoices, key figures maintain only statistical summaries, such as the total number of orders placed by a customer. This is one level of aggregation of transaction data, and it is *qualitative* in nature because the selection of key figures is subjective. Finally, data are collected or aggregated for specified time periods, such as daily, weekly, and monthly, which are specified in the **period definition**. In contrast to key figures, period definition represents a *quantitative* aggregation of data. Thus, information structures can be defined as aggregated and summarized forms of transaction data that are periodically updated.

In sum, then, the OLTP environment provides reporting in the form of lists, and the OLAP environment provides reporting in the form of analytics via information systems. Going further, lists fall into two categories—work lists and online lists.

WORK LISTS

Work lists identify tasks that are scheduled to be completed in a process. In the fulfillment process, for example, once customer orders are recorded in the system by the salespersons, warehouse personnel can retrieve a list of orders that are ready for picking. Picking is the first step in preparing a shipment. It involves retrieving, or picking, materials from storage. A *picking due list* (Figure 2-14) identifies all customer orders that must be prepared for delivery so that they can be shipped in a timely manner. The user selects the appropriate data, such as shipping location, due dates, and other relevant parameters (see inset in Figure 2-14). The resulting work list shows three orders that must be picked to ensure that they will be delivered on time. The user will select one of the orders and complete the picking task.

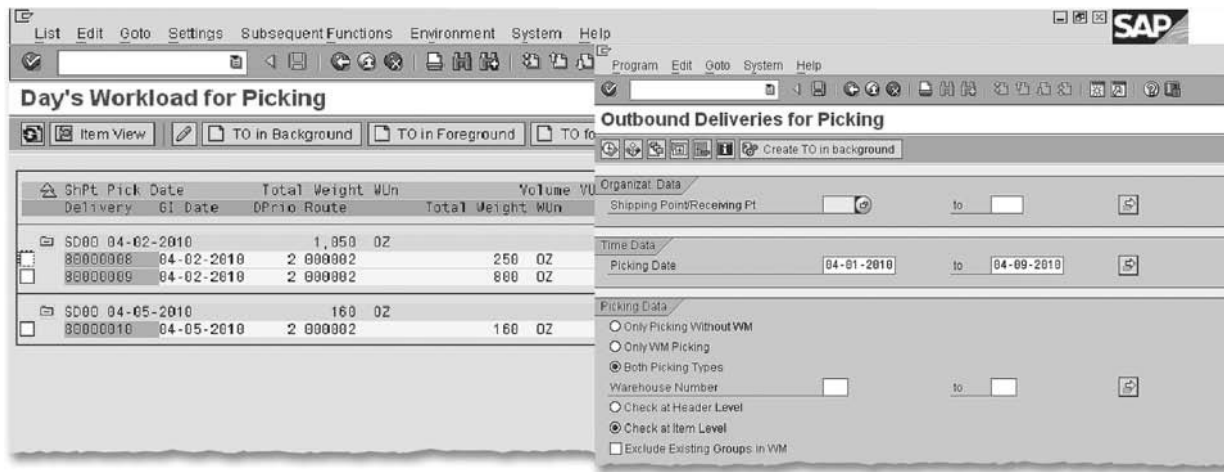


Figure 2-14: Work list—picking due list. Copyright SAP AG 2011

Demo 2.4: Review a work list

ONLINE LISTS

Online lists display lists of master data—such as materials, vendors, and purchasing info records—and documents—such as transaction documents, FI, CO, and material documents—that are generated during the execution of a process. The content and appearance of these lists are defined using selection parameters and scope-of-list parameters. *Selection parameters* determine which documents will be included in the list, while the *scope-of-list parameters* define which data will be included for the selected documents. In addition, numerous other parameters such as dates, master data (e.g., vendors), and organizational data (e.g., plant) can be used to narrow the data in the report. Figure 2-15 shows the initial screen for the display of purchasing documents. The inset shows choices for the selection parameters.

The reports are displayed in one of two available formats—a standard list format using the *SAP list viewer* or a grid format using the *ABAP list viewer (ALV) grid control*. Examples of these formats are provided in Figure 2-16 and Figure 2-17, respectively.

The list viewer and grid control provide several options for displaying the report output. These options are illustrated in Figure 2-18 and are explained as follows.

- The *select detail* allows the user to select a line on the report and drill down for more details.
- The *set filter* option is used to limit the list to only those items that meet specified values. For example, the list can be restricted to purchase orders where the quantity ordered (one of the columns) exceeds a specified value.
- Lists can be *sorted* in ascending or descending order in a selected column.

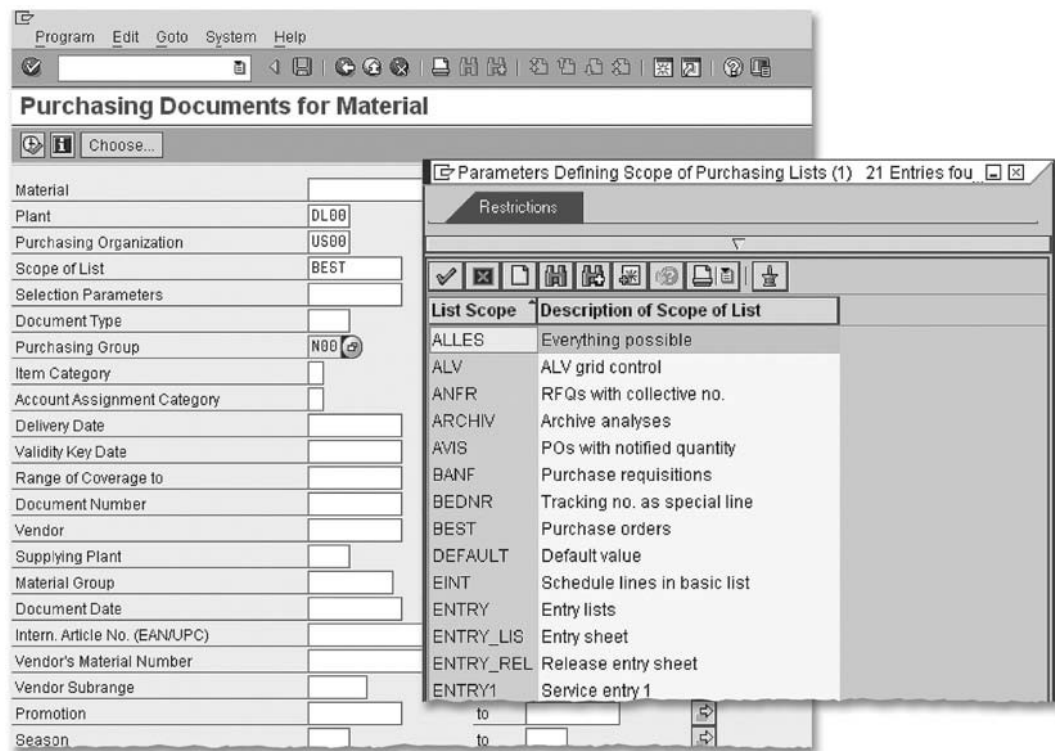


Figure 2-15: Online list—list display for documents. Copyright SAP AG 2011

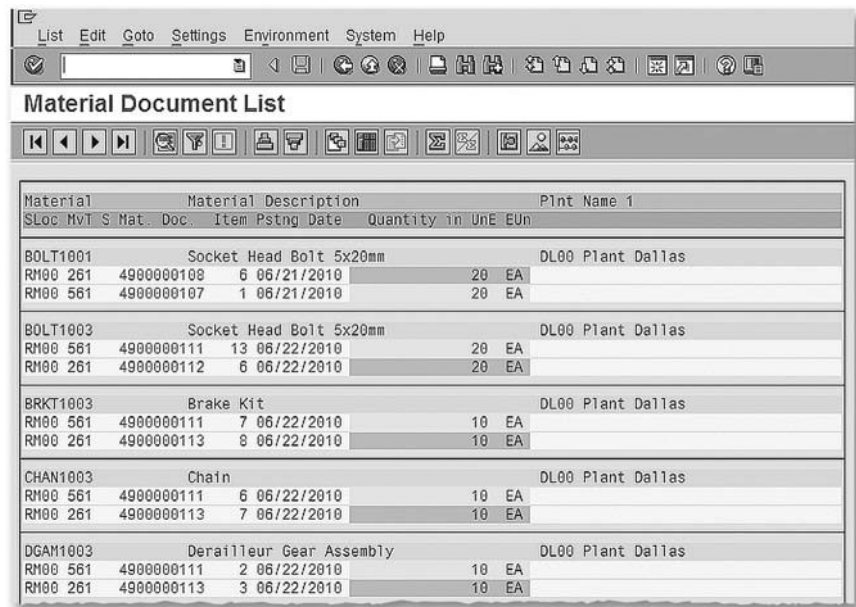


Figure 2-16: Online list—report using SAP list viewer. Copyright SAP AG 2011

Material Document List

Material	Plant	SLoc	MVT	S	Material Doc.	Item	Posting Date	Qty in Un. of Entry	EUn
BOLT1001	DL00	RM00	261		4900000108	6	06/21/2010	20	EA
BOLT1001	DL00	RM00	561		4900000107	1	06/21/2010	20	EA
BOLT1003	DL00	RM00	561		4900000111	13	06/22/2010	20	EA
BOLT1003	DL00	RM00	261		4900000112	8	06/22/2010	20	EA
BRKT1003	DL00	RM00	561		4900000111	7	06/22/2010	10	EA
BRKT1003	DL00	RM00	261		4900000113	8	06/22/2010	10	EA
CHAN1003	DL00	RM00	561		4900000111	6	06/22/2010	10	EA
CHAN1003	DL00	RM00	261		4900000113	7	06/22/2010	10	EA
DGAM1003	DL00	RM00	561		4900000111	2	06/22/2010	10	EA
DGAM1003	DL00	RM00	261		4900000113	3	06/22/2010	10	EA
DGAM9999	DL00	RM00	561		4900000100	1	06/21/2010	5,000	EA
DXTR3003	DL00	FG00	101		5000000005	1	06/22/2010	10	EA
HXNT1001	DL00	RM00	261		4900000108	4	06/21/2010	20	EA
HXNT1001	DL00	RM00	561		4900000104	2	06/21/2010	20	EA
HXNT1001	DL00	RM00	561		4900000104	1	06/21/2010	20	EA
HXNT1003	DL00	RM00	261		4900000112	4	06/22/2010	20	EA
HXNT1003	DL00	RM00	561		4900000111	11	06/22/2010	20	EA

Figure 2-17: Online list—report using ALV grid control. Copyright SAP AG 2011

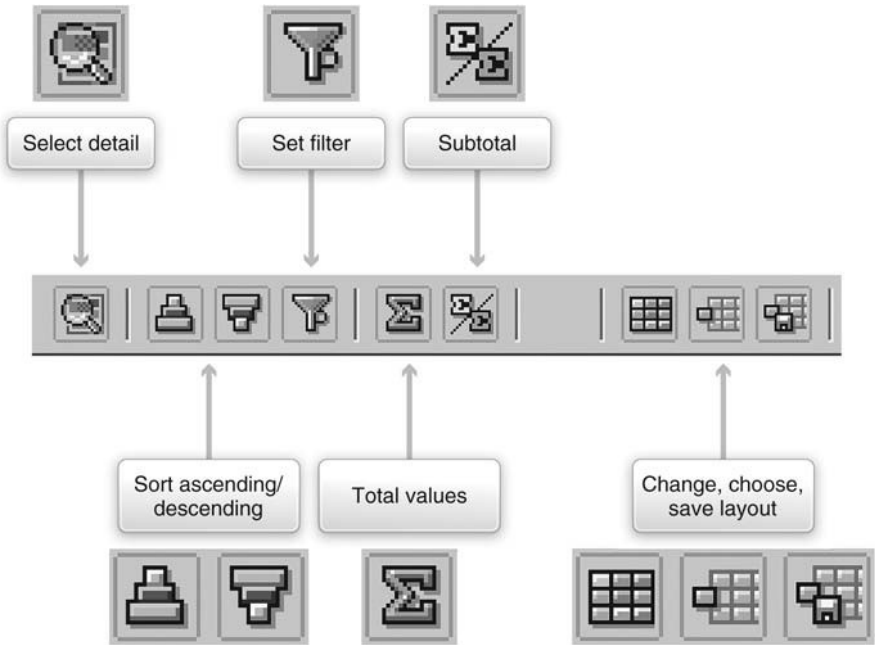


Figure 2-18: Functions of the list viewer and grid control

- The *total values* and *create subtotals* options enable users to calculate totals and subtotals for selected columns, respectively.
- The *layout* options allow users to change the appearance of the display. The layout option displays a list of available fields that can be included in the report. Fields appear as columns in the report and can be added or removed as desired. The modified layout can be saved as a *variant* of the report, which can be used to re-create the report at a later time.
- Several options are available to *export* the data in various formats, such as MS Word, Excel, and HTML. In addition, the data can be e-mailed using the system's messaging capabilities.

Demo 2.5: Review an online list

INFORMATION SYSTEMS

Numerous information systems are available within SAP ERP to support most of the processes that we will discuss in this book. Information systems (IS) can be divided into three broad categories: logistics information systems (LIS), financial information systems (FIS), and human resources information systems (HRIS). Logistics information systems support all of the logistics processes. Recall from Chapter 1 that logistical processes are concerned with acquiring, storing, creating, and distributing materials. The components of LIS include purchasing IS, sales IS, inventory control IS (IWM processes), quality management IS, plant maintenance IS, and shop floor IS (production processes). The financial information system supports reporting related to the general ledger (e.g., balance sheet, income statement, statement of cash flows), accounts receivable, and accounts payable. The HRIS is used to retrieve information about different HR components such as personnel, positions, and jobs.

Recall that information systems are part of the OLAP component of SAP ERP, and they utilize information structures to provide analytic capabilities. There are two types of information structures—standard and user-defined. **Standard information structures** are predefined in the SAP ERP system, and they collect the data needed to generate the most commonly used reports. SAP ERP also enables users to define their own structures, known as **user-defined information structures**, to meet specific reporting requirements.

Finally, information structures enable users to conduct two types of analysis—standard and flexible. **Standard analysis** provides predefined analytics for data in standard information structures. This type of analysis is sufficient for most analytic requirements. When standard analysis is insufficient, users can conduct flexible analysis to customize the reporting. In contrast to standard analysis, **flexible analysis** allows users to define the content and format of the analysis. Specifically, it enables users to combine available characteristics and key figures as needed and to create new key figures using user-specified formulas. It also provides users with several layout options.

To summarize, the OLTP component of SAP ERP offers reporting in the form of work lists and online lists that are based on detailed transaction data. In contrast, the OLAP component offers reporting via information systems,

based on aggregated data in information structures. These capabilities are limited to data within the SAP ERP system. Today, however, companies need even more powerful reporting capabilities that combine data from multiple sources. This is the domain of business intelligence, which we discuss next.

Demo 2.6: Review SAP ERP reports

BUSINESS INTELLIGENCE

Business intelligence is a general term that refers to the overall capabilities a company uses to collect and analyze data from a variety of sources to better understand its operations and make better managerial decisions. As we suggested in the previous paragraph, the OLAP environment within SAP ERP cannot perform the powerful analytic capabilities needed to provide business intelligence. For these purposes, businesses utilize SAP® Business Warehouse (SAP BW) (Figure 2-19).

SAP BW is a separate system that receives data from the SAP ERP system, other SAP systems such as SAP CRM and SAP SRM, and other non-SAP systems. These data are stored in the SAP BW database. Whereas the

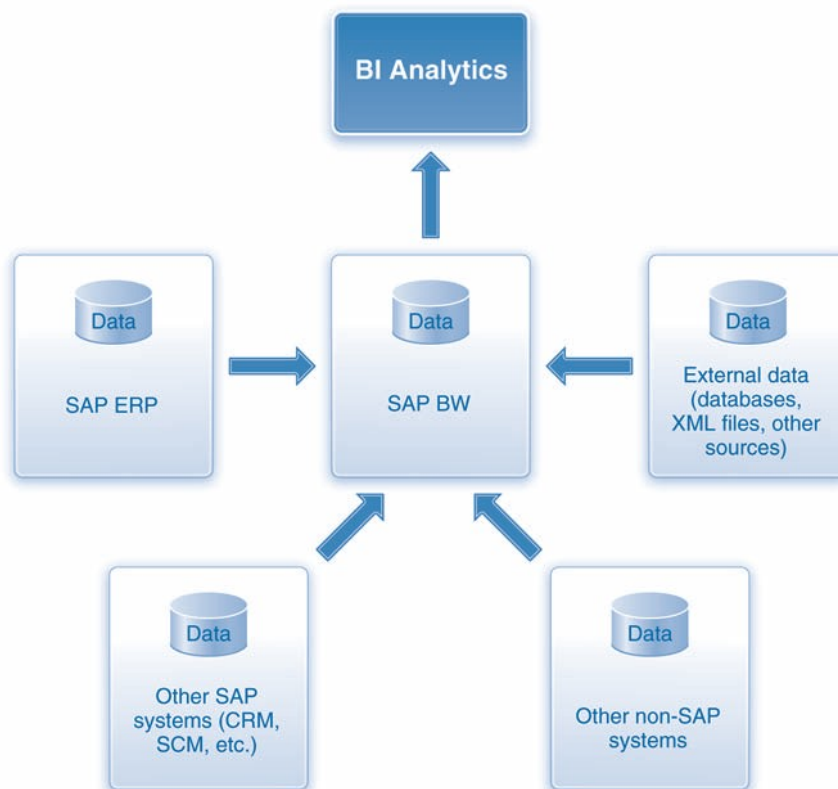


Figure 2-19: Reporting using SAP BW

SAP ERP system is used to execute process steps, the SAP BW system is designed and optimized for processing large quantities of data to provide powerful analytics. Unlike the OLAP environment in SAP ERP, SAP BW is not a real-time or an online system. That is, the data used in BW reporting are not tied to transactions and therefore are not the most current data available.

CHAPTER SUMMARY

This chapter explains the evolution of enterprise systems in terms of the processes they manage and the technical capabilities they possess. In addition, it examines the different types of data that ES collect and utilize to execute processes and enable managerial decision making.

Enterprise systems evolved from the custom-built mainframe applications of the 1960s and 1970s into the three-tier client/server systems of the 1990s. They were then rebuilt to take advantage of the new technical capabilities of service-oriented architectures to expand their reach and value as business platforms. The largest and most complex ES are integrated ERP systems. ERP systems initially managed only intra-company processes. However, their capabilities were later extended to incorporate external or inter-company processes, such as customer relationship management and supply chain management. The combined collection of inter-company process-based applications and intra-company process-based systems forms an ES “application suite.”

Enterprise systems generate and consume vast quantities of different types of data in their operations. Data in enterprise systems can be classified into three major categories: organizational data, master data, and transaction data. Organizational data represent the structure of the enterprise. Examples are company code, plant, and storage location. Master data represent the various entities or materials that are associated with processes. Materials master data contain information needed to procure, store, manufacture, ship, and invoice physical goods and services. Transaction data are collected during the execution of a process and contain information such as dates, quantities, prices, and payment and delivery terms.

Reports include organizational data, master data, and transaction data and are used to support process execution and for managerial decision making. ERP systems provide multiple options for extracting and analyzing the different types of data for different purposes. Information systems (IS) combine the various types of data for specific functions in the enterprise. IS also enable employees to easily access relevant data in ways that facilitate efficient decision making and operational oversight.

KEY TERMS

Application layer	Online lists
Application platforms	Online transaction processing (OLTP)
Application suite	Organizational data
Architecture	Period definition
Business intelligence	Plant
Characteristics	Presentation layer
Client	Product lifecycle management (PLM)
Company code	Raw materials
Customer relationship management (CRM)	Reporting
Data layer	Semifinished goods
Enterprise resource planning	Service-oriented architecture (SOA)
Financial accounting (FI) documents	Standard analysis
Finished goods	Standard information structures
Flexible analysis	Supplier relationship management (SRM)
Information structures	Supply chain management (SCM)
Key figures	Three-tier client-server architecture
Management accounting or controlling (CO) documents	Trading goods
Master data	Transaction data
Material documents	Transaction documents
Material group	User-defined information structures
Material types	Work lists
Online analytic processing (OLAP)	

REVIEW QUESTIONS

1. Describe client-server and services-oriented architectures. What are the advantages and disadvantages of each architecture?
2. What is an enterprise system application suite? Describe the capabilities of the individual components of the application suite.
3. Discuss the three types of data in an enterprise system and how they are related.